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TJA1121 Application note

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Application note
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Document information

Information	Content
Keywords	Automotive Ethernet, IEEE 1000BASE-T1, PHY, TJA1121A, TJA1121B, ASIL-B, MACsec
Abstract	The TJA1121 is a 1000BASE-T1 compliant Ethernet PHY with MACsec. The device provides 1000Mbit/s transmit and receive capability over twisted-pair cables, supporting a cable length of at least 15 m. It is available in two variants. The TJA1121A offers RGMII(-ID) MAC interface. The variant TJA1121B provides SGMII.

1 Introduction

The TJA1121 is a 1000BASE-T1 compliant Ethernet PHY. It provides 1 Gbit/s transmit and receive capability over shielded or unshielded twisted-pair cables, supporting a cable length of up to 15 m. The TJA1121A variant includes an RGMII(-ID) MAC interface. The TJA1121B variant features an SGMII interface.

A MACSEC engine providing encryption, decryption, signing as well as signature checking is embedded in the PHY, supporting these operations at line rate (1000 Mbit/s).

For sustainable system efficiency and flexibility, the device was optimized for lowest power consumption and highest robustness, while retaining a small package footprint.

The TJA1121 was developed in compliance with ISO26262, achieving ASIL-B. Adequate safety features have been implemented on-chip to support functional safety concepts at a system level. Additional documentation, including a safety manual, is available on request. With state-of-the-art diagnostic features, the device is perfectly suited to support high availability of communication paths in upcoming automotive electrical/electronic (E/E) architectures.

The TJA1121 is designed to meet the harsh requirements of automotive applications. It is typically used in electronic control units (ECUs) like zonal aggregators, gateways, telematics and head units, as well as typical end nodes like radar, LiDAR, and camera sensors for driver assistance systems (ADAS).

The device supports OPEN Alliance TC-10 sleep/wake over data-line. By eliminating the need for extra cables, the feature saves on cost and weight.

Due to its efficient IEEE1588v2 and 802.1AS-2020 compliant time stamping capability, the TJA1121 provides high-resolution accuracy for global time synchronization, which is required to support the timing critical data fusion functions of self-driving cars.

For more information, please visit https://www.nxp.com/TJA1121 for the TJA1121 data sheet.

2 1000BASE-T1 Basics

2.1 System diagram

The 1000BASE-T1 compliant Ethernet PHY provides 1000 Mbit/s transmit and receive capability over a single twisted-pair cable. The device supports a cable length of at least 15 m. It is optimized for capacitive signal coupling to the twisted-pair lines. A common mode choke (CMC) is typically inserted into the signal path to comply with stringent automotive EMC requirements. Standard RGMII interface (TJA1121A) or SGMII interface (TJA1121B) can be used to connect the PHY with a Media Access Control (MAC) unit.

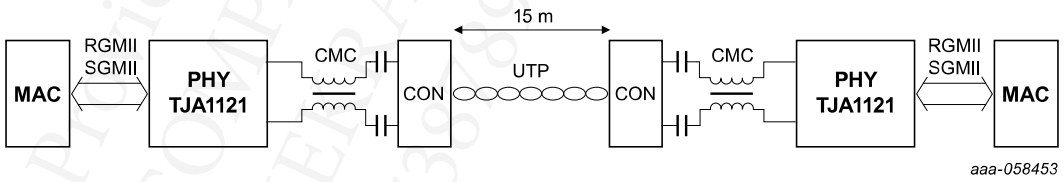


Figure 1. TJA1121 system diagram

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3 Hardware Application

3.1 Overview

A typical application circuit for the TJA1121 can be seen in Figure 2 for the A variant and in Figure 3 for the B variant. Details of the MDI interface are shown in Section 3.4. The application shown contains capacitive coupling to the twisted-pair wires. To comply with EMC requirements, a common-mode choke (CMC) should be inserted into the signal path. The coupling capacitors should be placed between the common mode choke and the connector.

A differential 100 Ω termination is integrated into the PHY. A common-mode termination network is recommended to avoid a floating common-mode behavior on the cable.

The TJA1121 requires three supply rails. There are minor differences between TJA1121A and TJA1121B. The IO voltage can be selected between 1.8 V, 2.5 V, or 3.3 V. Details are given in Section 3.5.

Control and status information is exchanged with the host controller via the SMI interface. The INH control output of the PHY can optionally be used to switch off the main voltage regulator when the PHY is in Sleep mode.

More detailed application information is provided in the following sections.

3.2 TJA1121A: RGMII interface

The TJA1121A supports RGMII with a number of submodes that can be selected via pin strapping or MDIO registers. The following list shows the available submodes. For more details, refer to Section 4.3 or the RGMII interfaces section of the TJA1121 data sheet [1].

Note: *RGMII modes listed in Table 1 are only available in the TJA1121A variant.*

Table 1. RGMII Modes

xMII mode	Short description
RGMII	RGMII 1.3 mode. Delays are expected on the PCB or the MAC side.
RGMII-ID (TXC)[1]	RGMII 2.0 mode. The TXC has internal delay enabled. The RXC has internal delay disabled.
RGMII-ID (TXC and RXC)	RGMII 2.0 mode. The TXC has internal delay enabled. The RXC has internal delays enabled.
RGMII-ID (RXC)	RGMII 2.0 mode. The TXC has internal delay disabled. The RXC has internal delay enabled.

[1] Not available via pin strapping

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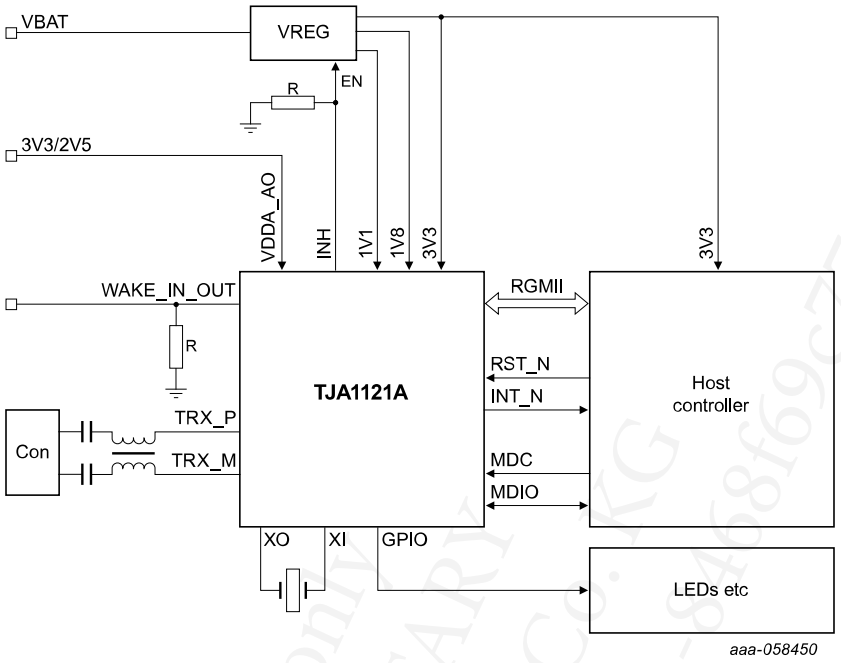


Figure 2. TJA1121A connection diagram

3.3 TJA1121B: SGMII interface

The TJA1121B uses a Serial-GMII (SGMII) SerDes interface for the PHY-MAC or PHY-PHY connection. Compared to RGMII, SGMII uses fewer interface signals. EMC performance is different due to smaller voltage swings and a changed coding scheme compared to RGMII.

Note: *SGMII modes listed in Table 2 are only available in the TJA1121B variant.*

Table 2. RGMII modes

xMII mode	Short description
SGMII PHY	1.25 GBaud SGMII mode with autonegotiation enabled as expected from MAC side.
SGMII MAC	1.25 GBaud SGMII mode with autonegotiation enabled. Received autonegotiation data is ignored.

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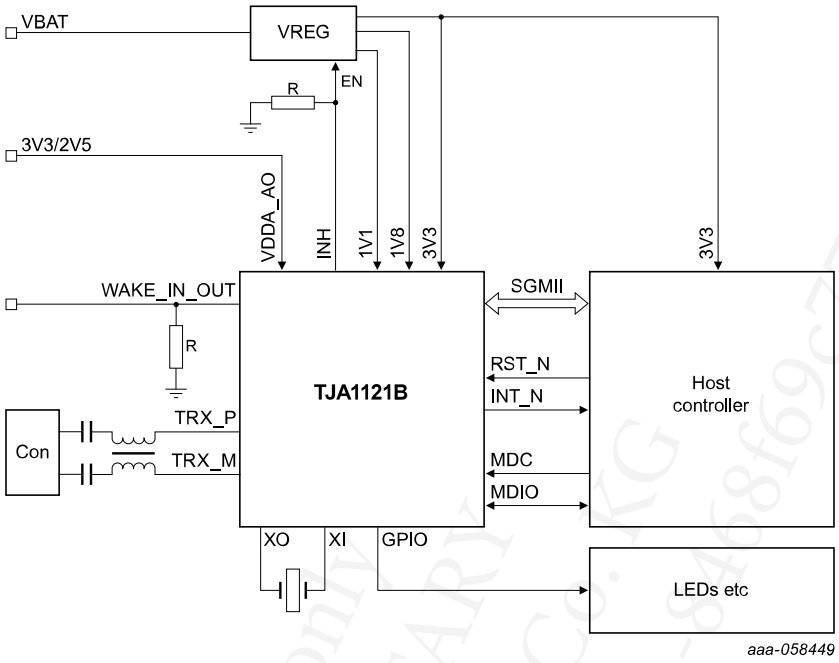


Figure 3. TJA1121B connection diagram

The TJA1121B provides the SGMII 4-wire interface which implements clock-data recovery (CDR) so no additional clock pairs are required.

The baud rate of SGMII is fixed at 1.25 Gbaud.

SGMII must always be AC coupled with a capacitor of 100 nF. The TJA1121B is compliant with the SGMII 1.8 specification. The SGMII interface implements (optional) Auto-Negotation. In this mode, PHY and MAC negotiate the supported interface capabilities to find the optimal operating conditions. In SGMII-PHY mode, the device communicates link speed and link availability by passing tx_config_Reg from PHY to MAC.

The TJA1121B automatically enables the SGMII transmitter (SON, SOP) on device startup in order to reduce time to communication. It is recommended to monitor the SGMII link status through COMPOSITE_STATUS. REPORTED_LINK_STATUS (30.8021h.15) register. With the LINK_STATUS_SELECT bits in the same register, the reported link status can be specified in more detail.

3.4 MDI interface

The MDI application circuitry is shown in Figure 4. The common-mode termination depends on OEM requirements and might vary for different OEMs.

The common-mode choke is expected to be compliant with the CMC specification given in IEC 62228-5 [3]. The 100 nF coupling-capacitors must have max. 10 % tolerance and ≥50 V voltage rating. The optional ESD suppression device is expected to be compliant to the ESD suppression device specification given in IEC 62228-5 [3].

For further information on the MDI interface, refer to the PCB design guidelines [2]

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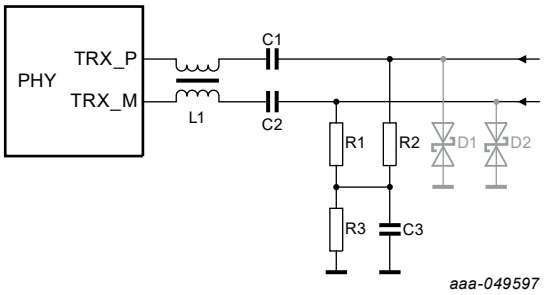


Figure 4. MDI diagram

Table 3. MDI Component specifications

Component	Parameter	Value	Unit
C_1, C_2	Capacitance	100	nF
	Tolerance	≤ 10	%
	Voltage	≥ 50	V
R_1, R_2	Resistance	1	kΩ
	Tolerance	≤ 1	%
	Power rating	≥ 0.4	W
C_3	Capacitance	4.7	nF
	Tolerance	≤ 10	%
	Voltage	≥ 50	V
R_3	Resistance	100	kΩ
	Tolerance	≤ 10	%
	Power rating	≥ 0.1	W
L_1	compliant to IEC 62228-5[1]		

[1] Function of DLW32MH101XT2 as well as ACT1210G-800-2P-TL05 has been confirmed in external compliance tests.

3.5 Supply

The TJA1121 requires supply from a 1.1 V, a 1.8 V, and a 2.5 V or 3.3 V rail. In addition, the IO voltage can be selected between 3.3 V/2.5 V/1.8 V. The following sections highlight the supply options.

Figure 5 shows the supply options for TJA1121A and Figure 6 the ones for TJA1121B. Ferrite beads and capacitors must not be merged into fewer parts. All 100 nF capacitors should be as close to the device as possible. The same is true for the ferrite beads. The pairs of ferrite beads and 1 µF capacitors shall be placed as close as possible.

Table 4 shows the required voltage for every supply pin. As some pins accept multiple voltages, the optimal or minimal supply rail configuration depends on the application.

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Table 4. Power consumption

Disclaimer: All values are taken from the data sheet. In case of contradictions data sheet values have precedence.

Device/Mode	Parameter	VDD_CORE	VREGA_IN	VDDA_RX	VDDA_AO	VDDA_TRX	VDDIO	VDDSI	VDDA_SO	Total
TJA1121A										
Typical	voltage [V]	1.1	1.8	1.8	3.3	1.8	1.8	-	-	
	current [mA]	290	1.5	95	1	30	16	-	-	
	power [mW]	319	2.7	171	3.3	54	28.8	-	-	**578.8**
Maximum	voltage [V]	1.1	1.8	1.8	3.3	1.8	1.8	-	-	
	current [mA]	380	2.5	110	2	35	60	-	-	
	power [mW]	418	4.5	198	6.6	63	108	-	-	**798.1**
Peak	voltage [V]	1.1	1.8	1.8	3.3	1.8	1.8	-	-	
	current [mA]	380	2.5	110	2	35	60	-	-	
	power [mW]	418	4.5	198	6.6	63	108	-	-	**798.1**
TJA1121B										
Typical	voltage [V]	1.1	2.5	1.8	3.3	1.8	1.8	1.1	1.1	
	current [mA]	290	7	95	1	30	2	20	13	
	power [mW]	319	17.5	171	3.3	54	3.6	22	14.3	**604.7**
Maximum	voltage [V]	1.1	2.5	1.8	3.3	1.8	1.8	1.1	1.1	
	current [mA]	380	8.5	110	2	35	5	24	15	
	power [mW]	418	21.3	198	6.6	63	9	26.4	16.5	**758.8**
Peak	voltage [V]	1.1	2.5	1.8	3.3	1.8	1.8	1.1	1.1	
	current [mA]	380	8.5	110	2	35	5	24	15	
	power [mW]	418	21.3	198	6.6	63	9	26.4	16.5	**758.8**

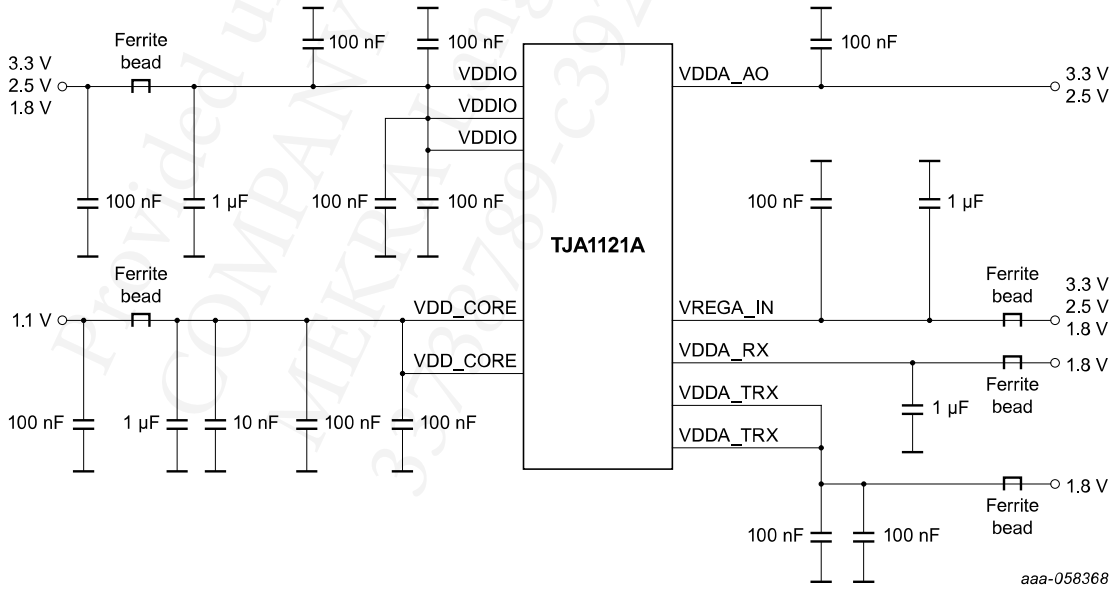


Figure 5. TJA1121A supply circuit with additional always-on supply

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Note: *As shown in Table 4 compared to TJA1121B the TJA1121A allows supplying VREGA_IN with 1.8 V. In this case it is possible to omit one of the ferrite beads and connect the nets supplying VREGA_IN and VDDA_RX/VDDA_TRX.*

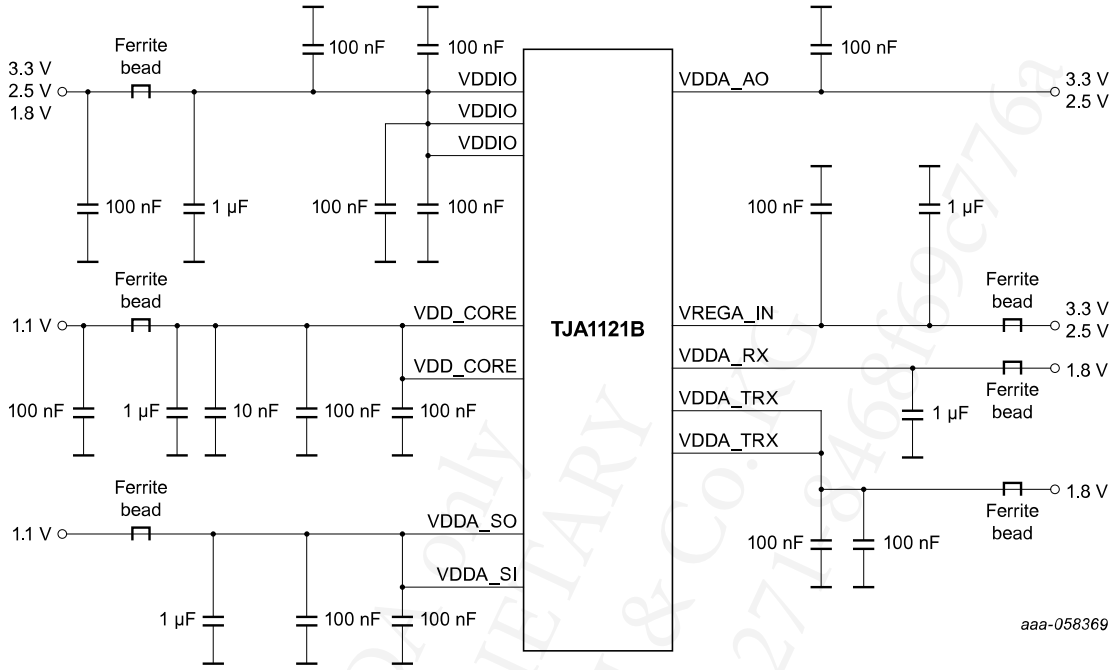


Figure 6. TJA1121B supply circuit with additional always-on supply

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3.5.1 Power-up/down and allowed supply ripple conditions

The TJA1121 has been designed in a way that no dedicated supply sequence is necessary. For instance, it does not matter in which order digital, analog and IO supplies are started. Being optimized for automotive use cases, the TJA1121 can also handle a wide range of slow and fast ramping supply conditions. See data sheet [1] for details.

Notes:

- Applying VDDIO before VDDA_AO is permitted but will result in a leakage current on VDDIO of approximately 11mA
- VDDA_RX and VDDA_TRX must be applied at the same time, i.e., it is expected bith pins are connected to the same supply rail.

Table 5. Recommended ripple per VDD supply

VDD supply	Conditions	Dynamic variation/ripple
3V3 VDDIO		< 50 mV(p-p)
2V2 VDDIO		< 35 mV(p-p)
1V8 VDDIO		< 25 mV(p-p)
VDD33_AO		< 50 mV(p-p)
VREGA_IN	analog supply	< 20 mV(p-p) @ f < 500 kHz
		< 10 mV(p-p) @ f ≥ 500 kHz
VDDA_RX	1.8V analog supply for 1000BASE-T1	< 20 mV(p-p) @ f < 10 MHz
		< 10 mV(p-p) @ f ≥ 10 kHz
VDDA_TRX		< 20 mV(p-p) @ f < 10 MHz
		< 5 mV(p-p) @ f ≥ 10 kHz
VDD11_CORE	1V1 supply	< 20 mV(p-p)
VDD_CORE	1V1 supply	< 20 mV(p-p)
VDDA11_SGMII	1V1 analog supply	< 20 mV(p-p) @ f < 500 kHz
		< 10 mV(p-p) @ f ≥ 500 kHz

Note: *Although no specific power sequencing is required when applying VDDIO before VDDA_AO leakage current of approximate 11mA will flow on VDDIO.*

3.5.2 Component specifications

Table 6 lists specifications for supply filtering components used in Figure 5 and Figure 6.

Table 6. Supply component specifications

Component	Parameter	Value	Unit
100 nF	Tolerance[1]	50	%
	Voltage	≥ 6.3	V
1 µF	Tolerance	50	%
	Voltage	≥ 6.3	V
10 µF	Tolerance	50	%
	Voltage	≥ 6.3	V

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Table 6. Supply component specifications...*continued*

Component	Parameter	Value	Unit
Ferrite bead[2]	impedance at 100 MHz	300	Ω
	DC resistance	≤ 0.05	Ω
	Current rating	≥ 400	mA

[1] Over lifetime, including dcomponent variation, temperature, aging, etc.
[2] Resistance and current parameters are dictated by the highest current, which only occurs on VDD_CORE. Less strict values are possible for ferrite beads on other supply rails.

3.6 Clocking

TJA1121 requires one of the following options:

- External 25 MHz ±100 ppm crystal. (XO mode)
- 1.8 V to 3.3 V single-ended 25 MHz clock
- Low-voltage differential 25 MHz clock
- 1.1 V single-ended 25 MHz clock[1]

Note: *When choosing one of the single-ended clock options, make sure to set pin strapping accordingly as supplying 3.3 V to XI while another option is selected may damage the device. This is not relevant for differential or XO modes.*

3.6.1 Crystal clock

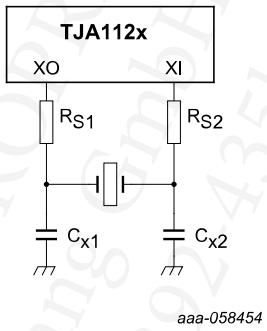


Figure 7. Crystal clock

Table 7. clocking circuit component example

Component	Parameter	Value	Unit	
R_{S1} / R_{S2}	Resistance	0	Ω	Not required. Can be added to limit drive strength. Align with crystal manufacturer.
C_{x1} / C_{x2}	Capacitance	10[1]	pF	Depends on PCB and crystal load capacitance. Align with crystal manufacturer.

[1] Recommended when using XRCGE25M000FXA1BR0

The following requirements must be fulfilled by the crystal:

1 This mode cannot be selected via pin strapping and is only available via register setting.

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Table 8. XTAL requirements

Parameter	Requirement
Frequency	25 MHz
Tolerance	50 / 100 PPM [1]
ESR	Max 100 Ω

[1] Over lifetime and operating conditions

3.6.1.1 Load capacitance and drive level

Every oscillation mode of the crystal is represented by an inductance, capacitance, and resistance in series. The fundamental frequency is represented by L_1, C_1 and R_1. All series branches are shunted by the parallel package capacitance C_p.

The equivalence model of the crystal in application is shown in Figure 8.

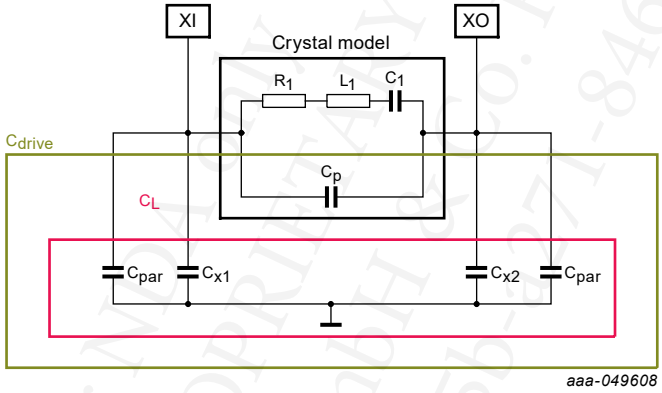


Figure 8. Crystal and application equivalence model

C_L is the typical load capacitance of the crystal and is specified by the crystal manufacturer. When using a crystal that is manufactured for a different load capacitance, the circuit oscillates at a slightly different frequency compared to the specified one.

The total load capacitance consists of PCB parasitics capacitance, the shunt capacitance of the crystal and the discrete load capacitors C_{x1} and C_{x2}.

$$C_L = C_{par} 2 + \frac{C_{x1} \cdot C_{x2}}{C_{x1} + C_{x2}}$$

The value of C_x can be calculated as follows:

$$C_{x1} = C_{x2} = (2 \cdot C_L) - C_{par}$$

The parasitics depend on the PCB layout and the TJA1121TJA1121's pin capacitance. As a rule of thumb, use $C_x = 2 \times C_L$ - 4 pF.

Note: *Always validate the PPM accuracy of the application. Ethernet demands a stable frequency with a ±100 PPM limits. The frequency is influenced by the board layout and selected components.*

The power dissipated in the crystal is called the drive level. In general crystal characteristics are specified at drive levels of 1 µW to 0.5 mW. Higher drive levels (between 1 mW and 5 mW) can accelerate aging or may change the crystal's physical parameters permanently. The level of drive can be calculated using equation:

$$P_{drive} = R_1 (2\pi f v_{eff} C_{drive})$$

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In which:

- R_1 is the series resistance of the crystal
- f is the oscillation frequency
- v_{eff} is the effective value of the voltage across the crystal, which is $1/\sqrt{2}$ of the amplitude.
- C_{drive} is the total differential capacitance $C_{drive} = C_L + C_P$.

The amplitude under typical conditions is 1.1 V.

3.6.2 Single ended clock

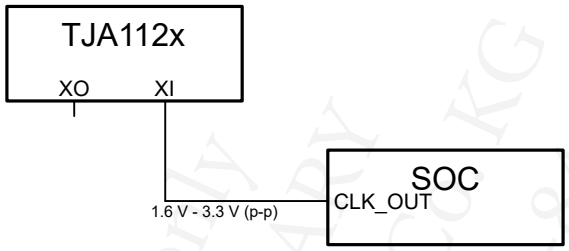


Figure 9. Single ended clocking input

Single ended clocking specification:

- Frequency: 25 MHz (TJA1121A: ±50 ppm, TJA1121B: ±100 ppm)
- Voltage: 3.3 V (0 to 3.6 V absolute, 1.6 to 3.6 V peak-to-peak)
- Shape: Square

Note: *In case a series capacitor is used in the clock signal, an external bias circuit is required to guarantee voltage levels within the limits for absolute voltage.*

3.6.3 Differential clock

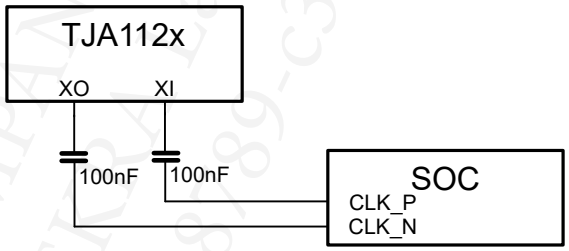


Figure 10. Differential clocking input

3.7 Host interface

Figure 11 shows a typical application. Care must be taken to keep the signals in the corresponding supply domain. Alternatively, use level shifters. Table 9 gives an overview of the relevant supply domains.

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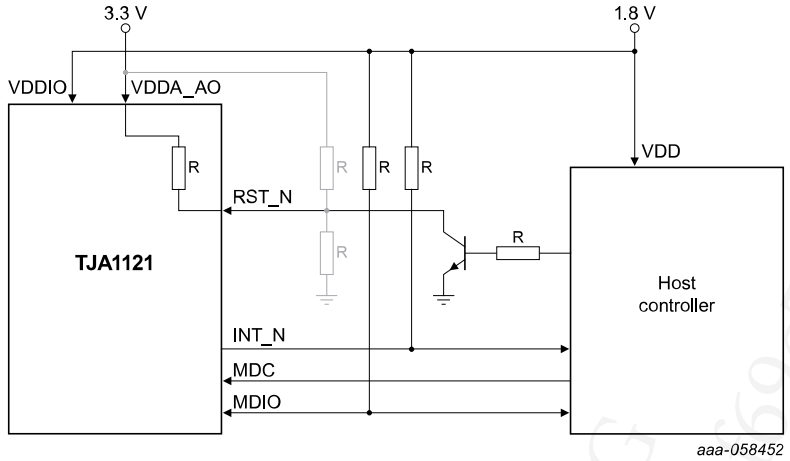


Figure 11. Host interface example VDDIO=1.8 V

Table 9. Supply domains

Supply domain	Contained pins
VDDIO	SMI (MDC, MDIO), INT_N, GPIO 0-11[1] RGMII (RXD0-3, RXC, RX_CTL, TXD0-3, TXC, TX_CTL)[1]
VDDA_AO	RST_N, WAKE_IN_OUT, INH

[1] Not all pins are available, depending on configuration.

3.7.1 Reset and Interrupt

The reset (RST_N) is an input signal to the PHY and typically driven by a GPIO pin of a microcontroller. The interrupt (INT_N) is an output of the PHY and an input signal to the microcontroller.

The interrupt pin (INT_N) is an open collector output. If multiple PHYs are used, INT_N can be shared and connected to a single interrupt input of the microcontroller. An external pull-up of 10 kΩ is needed. If the interrupt is unused, the INT_N pin can be left open.

The reset pin (RST_N) has an internal pull-up of 40-60 kΩ and no external pull-up or capacitor is needed. However, an additional external pull-up or pull-down can be applied in case a certain level is expected as default. The resistor value should be chosen such that a proper high/low level is ensured. This must account for the internal pull-up and other resistors in this line. For example, internal pull-up/pull-down in the microcontroller. See Figure 11 for details.

If the device shall be kept in reset on power-up, it is recommended to add a pulldown resistor of 10 kΩ or less.

RST_N logic levels are according to VDDA_AO.

3.7.2 SMI interface

The SMI (Serial Management Interface) is a serial bus defined in IEEE802.3 Clause 22 and Clause 45. SMI provides access to the various control and status registers for management and service purpose. The TJA1121 supports the Clause 22 protocol version as well as the Clause 45 version. However, Clause 22 access is limited to some basic registers and indirect access to Clause 45 registers.

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SMI communication must only be executed with either Clause 22 or Clause 45 as operating both versions simultaneously[2] is not supported.

In order to access the PHY, the correct PHY address must be known. The PHY address is configured as part of the pin strapping selection and a hardcoded part of the application software.

SMI broadcast through PHY address 0 is disabled by default and can only be enabled via SMI configuration which in turn requires initial knowledge about the PHY address. A typical use case is configuring several PHYs in parallel or changing the operation state of several PHYs simultaneously.

If for some reason the PHY address is not known but only one TJA1121 is present on the bus, a search can be implemented by trying to read the PHY_IDENTIFIER on all possible addresses (1-27).

3.7.2.1 MDIO Pullup

The SMI specification requires a pullup resistor on MDIO. It needs to ensure proper high level when the bus is idle and after every transaction. The recommended value is 1.5 kΩ to 10 kΩ. It must be guaranteed that a high level is reached within one clock cycle. Using the equation for RC circuits (1) and inserting the required timing and threshold conditions (2 and 3) it can be solved for R (4). Table 10 shows some example values under various conditions.

$$U(t) = U_{max}\left(1 - e^{\frac{t}{RC}}\right) \quad (1)$$

$$t = \frac{1}{2}t_{cycle} - t_{setup} \quad (2)$$

$$U(t) = 0.7 \text{ x } U_{max} \quad (3)$$

$$R = \frac{\frac{t_{cycle}}{2} - 10ns}{\ln(0.3)C} \quad (4)$$

Table 10. MDIO pullup resistor configuration

MDIO capacitive load	MDC clock frequency	MDIO pullup value (maximum)	MDIO pullup value (recommended)	Comment
10 pF	f = 2.5 MHz t_{cycle}=400 ns	15.8 kΩ	10 kΩ	Clause 22 speed, short traces, only one PHY connected
100 pF	f = 2.5 MHz t_{cycle}=400 ns	1.5 kΩ	< 1.5 kΩ	Clause 22 speed, long traces and/ or more than one PHY connected
10 pF	f = 10 MHz t_{cycle}=100 ns	3.3 kΩ	1.5 kΩ	High speed, short trace length
50 pF	f = 10 MHz t_{cycle}=100 ns	600 Ω	1 kΩ	High speed, medium trace length **IDLE_PRECHARGE setting required. (see below)**

If the time constant defined by MDIO pullup resistor and trace capacitance cannot be guaranteed to be smaller than one clock cycle, by setting SMI_CONFIG.IDLE_PRECHARGE (MMD30.018Ah.13) the PHY will actively drive MDIO after each read transaction to ensure a correct high level.[3]

2 When switching from Clause 22 to Clause 45 or vice versa perform a hardware reset and stick to the same access scheme until the next hardware reset.

3 It is recommended to enable this setting only when needed.W

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3.7.2.2 IEEE 802.3 Clause 45

For a full Clause 45 register access, two transactions are required. First an address frame is sent to specify the MMD (MDIO Managed Device Number) and register address. Then, a data frame is sent to perform the data read or write.

In addition to this, there exists a read operation that increments the register address which allows subsequent reads without sending address frames. The address pointer is kept individual for each MMD.

Table 11. Management frame fields

Frame	PRE	ST	OP	PHYADDR	DEVAD	TA	Adress/Data	Idle
Address	1...1	00	00	PPPPP	EEEEE	10	AAAAAAAAAAAAAAAA	Z
Write	1...1	00	01	PPPPP	EEEEE	10	DDDDDDDDDDDDDDDD	Z
Read	1...1	00	11	PPPPP	EEEEE	Z0	DDDDDDDDDDDDDDDD	Z
Post-read increment address	1...1	00	10	PPPPP	EEEEE	Z0	DDDDDDDDDDDDDDDD	Z

- ST = Start-of-Frame = ‘00’ (fixed) to define Clause 45 data frames
- OP = Operation code (’1x’ = Read, ’0x’ = Write)
- PRTAD = 5-bit Port address (MSB first) = PHY Address
- DEVAD = 5-bit (MMD)
- ADDR/DATA = 16-bit Address or Register value (bit 15 first)

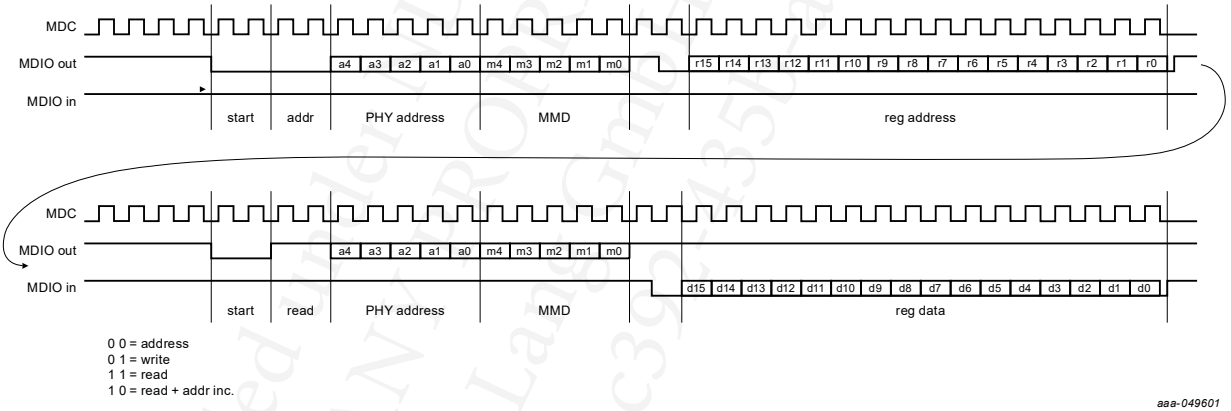


Figure 12. Clause 45 Read

In case multiple consecutive registers shall be read instead of repeating the address phase for every read, a read+address increment operation can be used. This doubles the throughput. An example is shown below.

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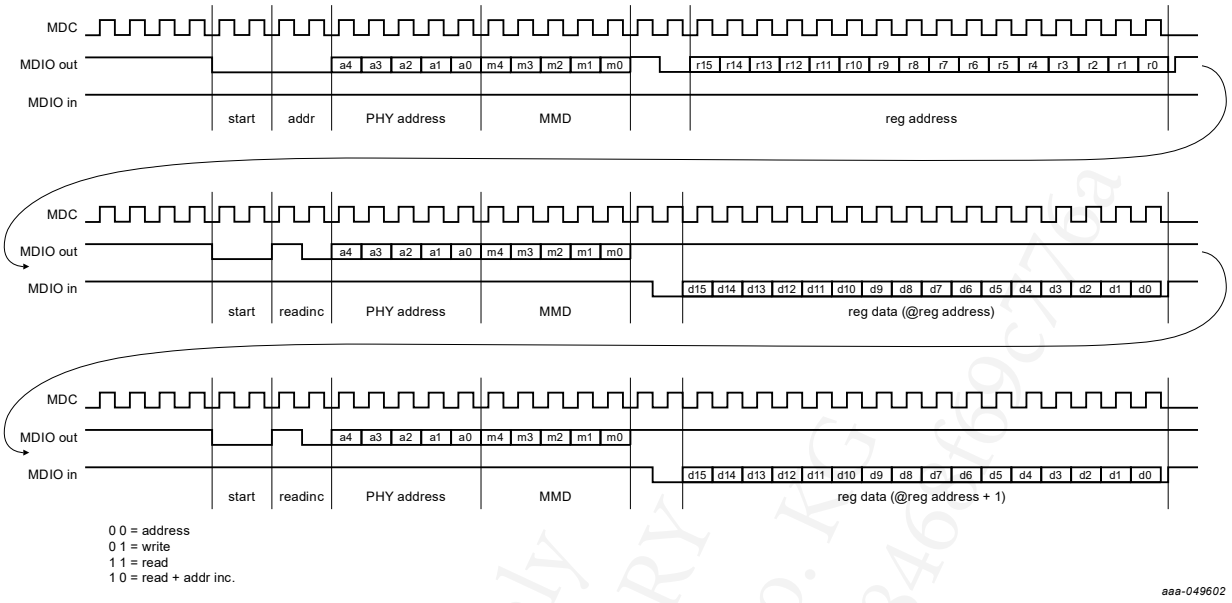


Figure 13. Clause 45 Read with address increment

3.7.2.3 IEEE 802.3 Clause 22

For Clause 22 register access, a single SMI frame specifies both address and data to be read or written. Figure 14 and Figure 15 show timing diagrams for read and write.

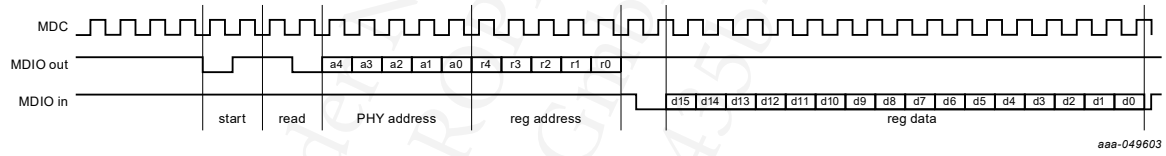


Figure 14. Clause 22 Read

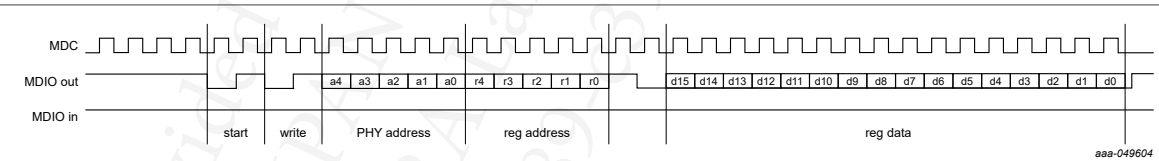


Figure 15. Clause 22 Write

Via Clause 22, the TJA1121 only allows direct access to 4 device registers as well as 3 additional registers that allow indirect access to Clause 45 registers.

The registers available under Clause 22 are:

- PHY_ID_1
- PHY_ID_2
- FAVORITE_IRQS
- ALWAYS_ACCESSIBLE

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- CL45_ACCESS_CONTROL[4]
- CL45_ADDRESS_DATA[5]
- CL45_ADDRESS[6]

FAVORITE_IRQS is a shortcut to various bits in the IRQ tree. All data presented here can also be found in the respective *_IRQ_MSTATUS register. ALWAYS_ACCESSIBLE is the only register accessible without the presence of VDD_CORE. It signals whether other registers are available for read and write and can be used to wake up the device from sleep. In addition, it can be used to read the status of various monitors.

The registers that allow access to Clause 45 registers are described in the Section 3.7.2.3.1.

Other registers reside in the Clause 45 space. In case the MAC does not support Clause 45, refer to Section 3.7.2.3.1.

3.7.2.3.1 Clause 22 to Clause 45 indirect addressing

There are registers in the Clause 22 space (CL45_ADDRESS, CL45_ADDRESS_DATA and CL45_ACCESS_CONTROL) which allow an indirect access scheme to read or write registers in the Clause 45 address space. For a complete Clause 45 transaction four steps must be performed:

1. Select MMD and set OP=address in CL45_ACCESS_CONTROL
2. Write Clause 45 register address to CL45_ADDRESS_DATA
3. Select MMD and set OP=data in CL45_ACCESS_CONTROL
4. Write or read register content to/from CL45_ADDRESS_DATA

Writes or reads to CL45_ADDRESS_DATA trigger the operation specified in CL45_ACCESS_CONTROL so the order of access must be followed.

The following listing shows an example reading PHY_CONFIG:

```
//write MMD=30+OP=address to CL45_ACCESS_CONTROL
write 13 0b00xxxxxxxx11110

//write PHY_CONFIG addr to CL45_ADDRESS_DATA
write 14 8108h

//write MMD=30+OP=data to CL45_ACCESS_CONTROL
write 13 0b01xxxxxxxx11110

//read data from CL45_ADDRESS_DATA corresponding to 30.8108h
read 14
```

In order to speed up the sequence shown a third register can be used to change the address without the need of changing the operation in CL45_ACCESS_CONTROL:

1. Select MMD and set OP=data in CL45_ACCESS_CONTROL
2. Write Clause 45 register address to CL45_ADDRESS
3. Write or read register content to/from CL45_ADDRESS_DATA

If subsequent accesses fall into the same MMD, the write to CL45_ACCESS_CONTROL can be omitted which reduces the number of clause 22 transactions from 4 to 2.

4 Used for clause 22 to clause 45 indirect addressing
5 Used for clause 22 to clause 45 indirect addressing
6 Used for clause 22 to clause 45 indirect addressing

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The following listing shows an example reading PHY_CONFIG followed by a write to another register in the same MMD

```
//write MMD=30+OP=data to CL45_ACCESS_CONTROL
write 13 0b01xxxxxxxxx11110

//write PHY_CONFIG addr to CL45_ADDRESS
write 30 8108h

//read data from CL45_ADDRESS_DATA corresponding to 30.8108h
read 14

//write a different addr to CL45_ADDRESS
write 14 8100h

//write data to the register
write 14 0bxxxxxxxxxx000100
```

3.7.2.4 Mixing Clause 22 and Clause 45 access

When the device is woken up under Clause 45 via the ALWAY_ACCESSIBLE register (address Shared.801Fh) and the next SMI access is performed under Clause 22, the wrong data will be returned.

Workaround: If clause 45 is used to wake up the device, make at least one more clause 45 access to a different register before switching to Clause 22. Switching from clause 22 to clause 45 is not affected.

3.8 WAKE_IN_OUT pin

The bidirectional WAKE_IN_OUT pin can be used as a local wake-up input or wake-up output. The wake-up output is intended to forward a wake-up to another PHY or switch. The input logic is positive. Thus, a rising edge triggers a wake-up event. The output logic is open-drain. When inactive the pin is floating. When generating a wake-up pulse, the pin is pulled to VDDA_AO.

An external pulldown resistor is required on this pin when it is being used (it must not be floating). A value between 10 kΩ and 100 kΩ is recommended. When not being used, it can be left unconnected.

The value of the resistor depends on the total capacitance and the desired fall time. The leakage current of the input is less than 1 µA and can be ignored. The input capacitance of the pin is less than 5 pF. If a level of 0.35×VDDIO is required as a low threshold level, a simple RC low-pass calculation can be used: Assuming a total capacitance of 50 pF and a desired fall time of 5 µs the resistor value must be 100 kΩ.

The actual behavior of the pin can be configured via MMD30.WISE_CONFIG.WU_IO (30.0182h 12:13). The following table shows the supported options:

Table 12. Wake-up IO configuration - WU_IO_ENABLE

Bits 12:13	Description
00	Wake-up IO is disabled
01	Wake-up IO is input only, output disabled
10	Wake-up IO is output only, input disabled
11	Wake-up IO is input and output

The length of a generated wake-up pulse can be configured via MMD30.WISE_PARAMETERS.WU_IO_PULSE_LENGTH.(30.0184h.3:0)

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3.9 INH pin

The intention of INH is to control voltage regulators within the ECU. In TJA1121 the voltage regulator for the host controller and the PHY is enabled via the INH output of the TJA1121.

The pin INH provides a VDDA_AO-related high-side switch which is floating during Sleep mode. This design allows combining INH/EN pins of multiple devices to achieve a wired-or topology. Thus, if another device is not actively driving INH high, a LOW signal on the inhibit input disables the regulator.

An external pulldown or a pulldown embedded in the voltage regulator is needed to ensure a low level. The recommended value is 5 kΩ to 20 kΩ. This is shown in Figure 16.

If the TJA1121 is not in sleep, the pin INH is switched to the VDDA_AO voltage. This indicates that the voltage regulator shall be enabled.

If the wakeup/sleep feature is unused and the INH pin is not used, it can be left open.

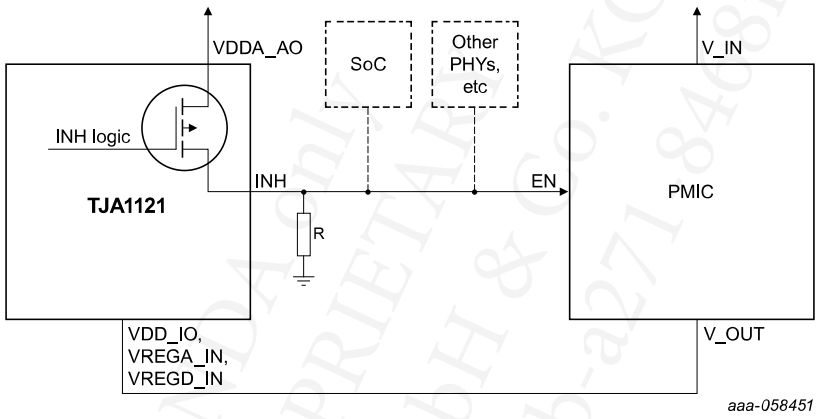


Figure 16. INH topology example

3.9.1 Voltage level adaption

In case the PMIC has a higher enable threshold than $V_{DDA(AO)}$ a voltage level translation must be implemented. Figure 17 shows an example circuit with component values listed below.

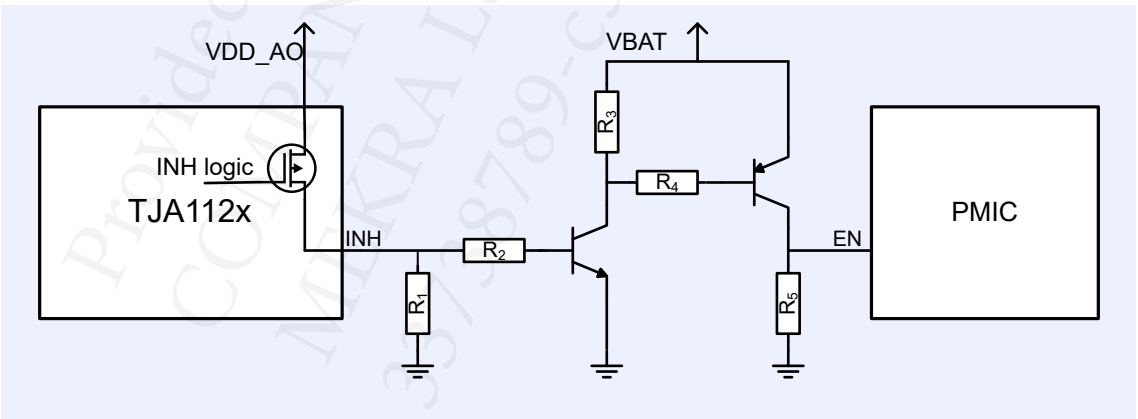


Figure 17. INH voltage level adaption

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Table 13. INH translation circuit component specifications

Component	Parameter	Value	Unit
$V_{DDA(AO)}$	Voltage	3.3	V
V_{BAT}	Voltage	12	V
R_1	Resistance	10	kΩ
R_2	Resistance	68	kΩ
R_3	Resistance	100	kΩ
R_4	Resistance	68	kΩ
R_5	Resistance	4.7	kΩ
Q_1	Type	BXC70C	
Q_2	Type	BCX71C	

3.10 Pin strapping

The TJA1121 allows for basic hardware configuration via pin-strapping during power-on. The following pins are used in several function blocks:

Table 14. Pinstrap pin to function mapping

Pin number	Name	Function
17	CONFIG0	PHY address
19	CONFIG1	
20	CONFIG2	
21	CONFIG3	RGMII/SGMII mode
23	CONFIG4	
24	CONFIG5	Clock mode
25	CONFIG6	1000BASE-T1 Leader/Follower, Autonomous

Each pin can have three states as listed in the table below. The pullup/down resistors should have a value of 10 kΩ. Other circuits connect to the pins must not interfere during the pin-strapping phase. Additional (parasitic) impedance that conflicts with the intended pin strap state must be avoided. See Section 3.10.7

Table 15. Three-level strap resistor Values

Setting name Abbreviation	Pullup resistor	Pulldown resistor
Open (O)	Open	Open
High (H)	10 kΩ	Open
Low (L)	Open	10 kΩ

The state of the pin strap is sampled at device startup or device reset. The result of the pin strapping is latched in registers accessible via SMI. All settings can be read and modified (except the PHY address) through MDI.

Note: *In case multiple TJA11xx PHYs are used on the same board CONFIG pins must not be tied together as the pinstrap logic of the PHYs might interfere and prevent correct detection.*

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3.10.1 Detection logic

The actual detection takes place in two phases:

1. Enable internal pulldown (disable pullup) and sample pin
2. Enable internal pullup (disable pulldown) and sample pin

The four possible results are decoded to the effective pin strap configuration as follows:

Table 16. Pinstrap decoding

Sample result pulldown phase	Sample result pullup phase	Decoded result	Pinstrap result
0	0	External pulldown	L
1	1	External pullup	H
0	1	Floating	O
1	0	Invalid	O[1]

[1] The startup is not interrupted so a valid result has to be assumed

Figure 18 shows a simplified version of the IO-cell. The colored boxes contain the resistors relevant for the three different configuration options. The value of the internal pullup/pulldown resistors is between 40 kΩ - 62 kΩ which leads to the following limits for the external resistors:

Table 17. Pinstrap constraints

Desired value	Detection threshold requirement	Constraint
H	V_{IO} > VDDIO×0.7	R_{pullup} < 17 kΩ
L	V_{IO} < VDDIO×0.3	$R_{pulldown}$ < 17 kΩ
O	V_{IO} > VDDIO×0.7 with internal PU V_{IO} < VDDIO×0.3 with internal PD	RP_{pullup} > 145 kΩ $RP_{pulldown}$ > 145 kΩ

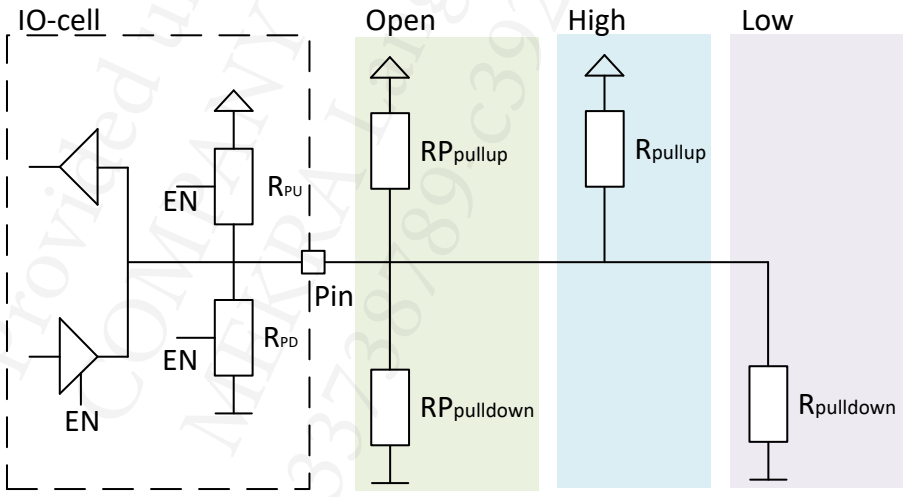


Figure 18. Simplified IO-Cell

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3.10.2 Pin strapping example

Figure 19 highlights several different pinstrap options. The shown setup results in the following configuration:

Table 18. pin-strapping example

Pin	Setting	Config
CONFIG0	High	PHYADDR=7
CONFIG1	Open	
CONFIG2	High	
CONFIG3	Open	xMII Mode = RGMII-ID
CONFIG4	Open	
CONFIG5	Open	Clock = differential
CONFIG6	Low	1000BASE-T1 = Follower/Non-autonomous

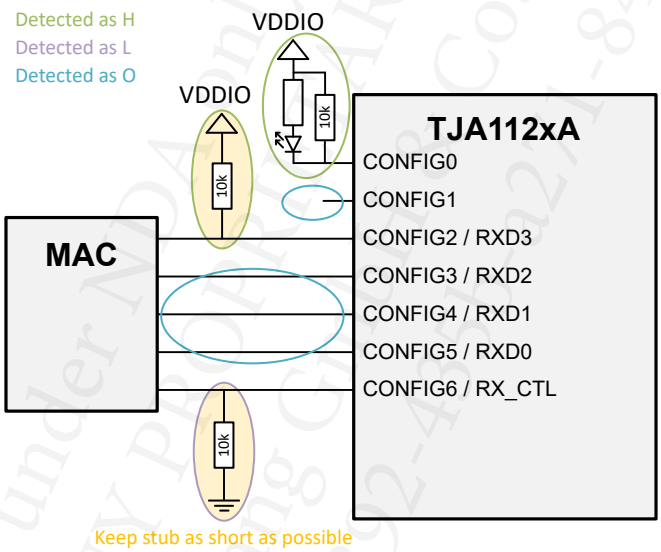


Figure 19. Pin strapping hardware

3.10.3 PHY Address Configuration

The PHY address is used for SMI addressing. . PHY addresses 1 to 27 can be selected via three pin strap pins. For details refer to the data sheet [1]. Figure 20 shows an example for PHY address = 2. As the PHY address is used in the SMI communication itself, it is not possible to change the PHY address via SMI access.

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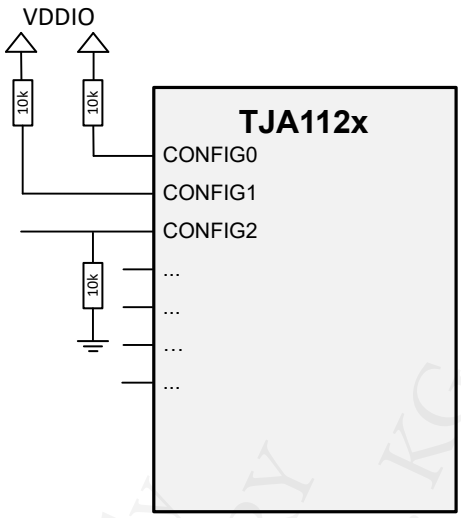


Figure 20. PHY Address config example

Table 19. PHY address pin-strapping

PHY address	Pin CONFIG0	Pin CONFIG1	Pin CONFIG2
1	H	H	H
2	H	H	L
3	H	H	O
4	H	L	H
5	H	L	L
6	H	L	O
7	H	O	H
8	H	O	L
9	H	O	O
10	L	H	H
11	L	H	L
12	L	H	O
13	L	L	H
14	L	L	L
15	L	L	O
16	L	O	H
17	L	O	L
18	L	O	O
19	O	H	H
20	O	H	L
21	O	H	O

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Table 19. PHY address pin-strapping...*continued*

PHY address	Pin CONFIG0	Pin CONFIG1	Pin CONFIG2
22	O	L	H
23	O	L	L
24	O	L	O
25	O	O	H
26	O	O	L
27	O	O	O

3.10.4 xMII Mode Configuration

Two configuration pins determine the default RGMII/SGMII mode. The possible options and corresponding pin strap settings are listed in Table 20. Pinstrap configuration, which does not correspond to valid xMII modes, are forbidden and should be avoided.

The TJA1121A supports CMOS-type RGMII modes. The TJA1121B supports SGMII modes.

Table 20. xMII pin-strapping

RGMII Mode (TJA1121A)	SGMII Mode (TJA1121B)	Pin CONFIG3	Pin CONFIG4
RGMII-ID (RXC)	SGMII-PHY	O	O
-	SGMII-MAC	O	H
-	-	O	L
RGMII		H	O
-	-	H	H
-	-	H	L
RGMII-ID (TXC/RXC)	-	L	O
-	-	L	H
JTAG	JTAG	L	L

3.10.5 Clock Configuration

Pin CONFIG5 controls the clock input settings.

Table 21. Clock pin-strapping

CLK select	Pin CONFIG5
differential	O
XO mode	H
single ended	L

Notes:

- *The selected clock mode impacts the maximum allowed voltage on the XI/XO pins. Only the single ended mode allows signal level up to VDDIO, other modes are restricted to 1.1 volts.*
- *A second single-ended mode, which allows a voltage swing of 1.1V, can be selected via register setting. Refer to Section 3.6 for more details.*

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3.10.6 Miscellaneous Configuration

Pin CONFIG6 controls the following miscellaneous settings.

- Leader/Follower
- Autonomous mode[7]

In case the TJA1121 is pin-strapped for Managed mode operation, the PHY pauses at startup, until the MMD30. DEVICE_CONTROL.START_OPERATION bit (30.0040h.0) is set. During this time, the desired Leader/Follower mode should be set.

Table 22. Misc pin-strapping

Leader/Follower	Autonomous Mode	Pin CONFIG6
Leader	enabled	O
Follower	enabled	H
register controlled (default: Follower)	disabled	L

3.10.7 Alternate Function Considerations

GPIO0/CONFIG0 is configured as LED per default. When connecting an LED, the polarity must be matched to the pin-strapping resistor as the LED acts as a resistor and overrule other pin-strapping configurations. Alternatively, a FET must be used to detach the load from the pin strapping mechanism.

GPIO2/CONFIG1 is configured as input per default. This results in a high-impedance state after the pin-strapping is completed. As the last state of the pin-strapping is high, a long discharging curve that exceeds the reset time might be visible.

CONFIG2 - CONFIG6 are part of the RGMII interface and configured as outputs after pin-strapping is completed. No undefined or intermediate states are visible on these pins.

3.11 IO pins

TJA1121 has 2 dedicated general purpose I/Os pins and, depending on the xMII configuration, up to 6 pins that can optionally be configured as GPIOs. Please refer to section 6.13.2 in the datasheet for details on which GPIO is available in which xMII mode and how to configure it.

The default configuration for all GPIOs is input with the exception of GPIO0 which is configured as link status LED. In order to manually configure a GPIO as input, FUNC_SELECT & SIGNAL_SELECT of the corresponding GPIOx_FUNC_CONFIG register must be set to MISC (3b111) and SNIF(5b00000) respectively.

3.12 JTAG

JTAG can be used to test all IOs available on the TJA1121.

In order to utilize JTAG the interface must be activated via pinstrapping. One solution is to let the host controller dynamically configure pinstrapping before switching the corresponding pins to xMII function.

Make sure the power supply for the TJA1121 is not controlled via INH when using JTAG because this pin will be switched off in the initialization of JTAG.

Please note that in case GPIO0, GPIO1, GPIO2 or GPIO9 are used for LEDs these will flicker while JTAG is active.

7 Compared to TJA1103/4 polarity correction cannot be set via pin-strapping. The default configuration is enabled, which can be changed via register configuration.

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Some JTAG pins are shared with pins used for pinstrapping. Therefore impact of both functions must be considered carefully. Known caveats are given in the following list:

- GPIO0/TMS/CONFIG0 will be pulled high during JTAG mode which will result in CONFIG0=H when device reset is triggered in JTAG mode. In case CONFIG0 is configured as High-Z the PHY-address will change during the reset process.

Table 23. JTAG pin conflicts

Symbol	Pin	Problem description	Solution / Workaround
GPIO0/TMS/ CONFIG0	17	The pin will be pulled high during JTAG mode which will result in CONFIG0=H when device reset is triggered in JTAG mode. In case CONFIG0 is configured as High-Z the PHY-address will change during the reset process.	• Ensure CONFIG0 is configured as (H)igh or (L)ow. This prevents the internal pull-up to change the pinstrapped PHY-address on device reset. • Sweep through all possible PHY addresses when discovering PHY. • Don't use SMI communication during JTAG testing.
		TMS pin of multiple PHYs might be connected in parallel. Pinstrapping operation of multiple devices will interfere with results. Implementation does not guarantee precise pinstrap timing which will lead to unpredictable results.	• Ignore values produced by CONFIG0. This means the PHY address can only be controlled reliably via CONFIG1 and CONFIG2 leaving only 9 out of 27 options. PHYs configured to a specific address can be expected on any of the 3 possible CONFIG0 values. E.g. PHY configured to address=1 can have adress=1, address=10 or address=19. • Connect TMS signals of multiple PHYs not directly on PCB but only via JTAG test fixture. This way the described issue only occurs as long as the JTAG test fixture is connected.
GPIO1/TDI	18	No known problems	
RX_ER/GPIO2/TDO/ CONFIG1	19	Pin is used in pinstrapping operation to set PHY-address. Next device in JTAG chain might interfere with pinstrapping.	• Ignore values produced by CONFIG1. See CONFIG0 description for details. • Connect JTAG signals only via JTAG test fixture. See CONFIG0 description for details.

4 Basic Control and Basic Status

4.1 PHY initialization

To get to a successful link-up an initialization is required which contains mandatory and optional steps. Some steps can be performed via pin-strapping while others require register writes. Figure 21 shows the required steps for initializing the PHY, highlighting mandatory items in red, leaving all other items as optional.

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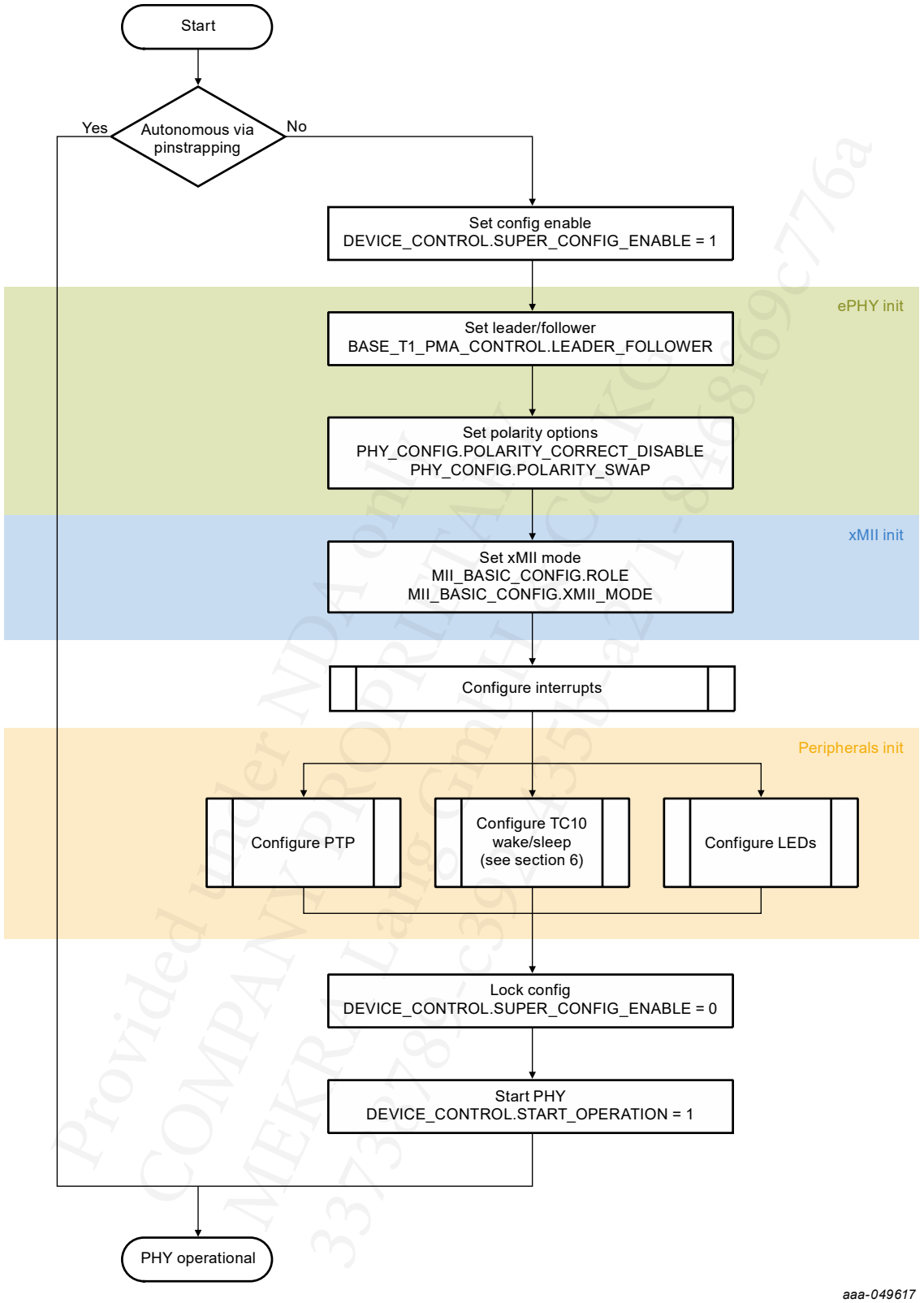


Figure 21. Example flow: Complete PHY initialization

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4.1.1 Mandatory steps

The following items must be configured for a functioning PHY.

- MDIO PHY address
- PHY mode: managed or autonomous
- VDD_CORE work-around settings
- PHY Leader/Follower mode
- xMII mode
- clock mode

The only configuration that can only be set via pin-strapping is the MDIO PHY address. It is required for all MDIO reads and writes and cannot be changed via register configuration.

The PHY can be configured via pin-strapping to start in autonomous or managed mode. As configuring the VDD_CORE work-around is mandatory before bringing up the link, it is recommended to configure the PHY to start into managed mode via pin-strapping. In managed mode, the PHY is booting into Standby mode and must be started manually. In autonomous mode the PHY completes the initialization process and switches into active mode. To be able to apply the VDD_CORE work-around, it needs to be put into standby mode again, potentially resulting in a short timespan where the PHY links up with undesired settings.

The VDD_CORE work-around settings are listed at the end of this section, see Table 25

Setting the PHY to Leader or Follower is mandatory, as a link can only be established between one Leader and one Follower, there is no autonegotiation available. In managed mode, it can only be set via register configuration.

xMII mode must be set to allow valid frame transmission between PHY and MAC. It can be either done via register configuration or via pin-strapping. Configuration via pin-strapping is recommended as most applications have a fixed xMII mode which is dictated by the hardware configuration.

Clock mode must be selected according to the type of clock signal provided. Similar to the xMII mode it can be configured via register or pin-strapping while pin-strapping is the preferred mode.

Note: *As the clock input might be damaged when a single ended clock is supplied while a low voltage mode is selected (differential or XO mode) make sure that this situation never occurs. When configuring clock input via register, it is recommended to select a single ended mode via pin-strapping and only switch to a low voltage mode once it can be guaranteed that no single ended signal is present.*

Table 24. Minimal initialization example

Action	MMD. Address	Value	Comment
COMMENT	Ensure pin-strapping: PHY-address, managed mode, clock mode, xMII mode		
INCLUDE	Insert VDD_CORE work-around from Table 25		
WRITE	30.0040h	2000h	MMD30.DEVICE_CONTROL.SUPER_CONFIG_ENABLE
WRITE	01.0834h	C001h	MMD1.BASE_T1_PMA_CONTROL.LEADER_FOLLOWER
WRITE	30.0040h	0001h	MMD30.DEVICE_CONTROL.START_OPERATION

4.1.2 Optional steps

In addition to the settings mentioned as mandatory, there are various items that can be configured depending on the applications needs. The most common ones are listed below. This list is not exhaustive as every register setting can be considered as an optional initialization item. Refer to the respective sections in Section 5 for more details.

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- Polarity correction / polarity swap
- Interrupt configuration
- LED configuration
- Function block configuration
  - PTP
  - Wake/Sleep

Table 25. VDD_CORE work-around register settings

Action	MMD. Address	Value	Comment
WRITE	30.01F8h	89C7h	1st step of enabling write
WRITE	30.01F9h	0893h	2nd step of enabling write
WRITE	30.0476h	58A0h	Modify the internal LDO settings
WRITE	1.8921h	0A3Ah	Modify the ADC latch configuration
WRITE	1.89F1h	16C1h	OPTIONAL; increased SQI averaging, see next topic "SQI averaging"
READ	30.0476h	58A0h (with mask FDFBh)	OPTIONAL; check for correct write command; **NOTE**: bit-2 & bit-9 are status bits and should be ignored for comparison
READ	1.8921h	0A3Ah	OPTIONAL; check for correct write command
READ	1.89F1h	16C1h	OPTIONAL; check for correct write command
WRITE	30.01F8h	0000h	1st step of disabling write
WRITE	30.01F9h	0000h	2nd step of disabling write

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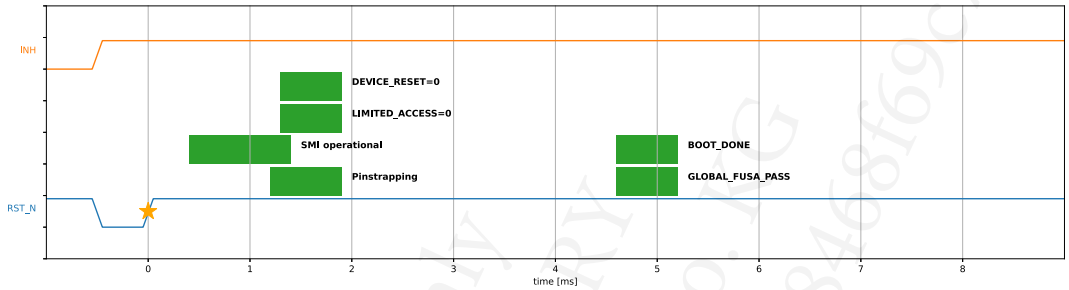
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4.1.3 Startup timing

The following charts show startup timings and notable events during bootup.

Description: *Level changes on external pins are depicted as logic level plots. Changes in observable PHY states (via register reads) are shown as green blocks highlighting measurement uncertainty. The end of a green block marks the point in time where the event will have taken place at the latest. All values are typical average values obtained through 100 iterations. Therefore all timings are indicative and should not be relied upon. The star marks the event that was used to trigger the process and start the timing measurement.*



Note that INH is asserted as soon as RST_N is asserted. In this chart this happens before 0ms.

Figure 22. Startup timing: Hardware reset / Power on

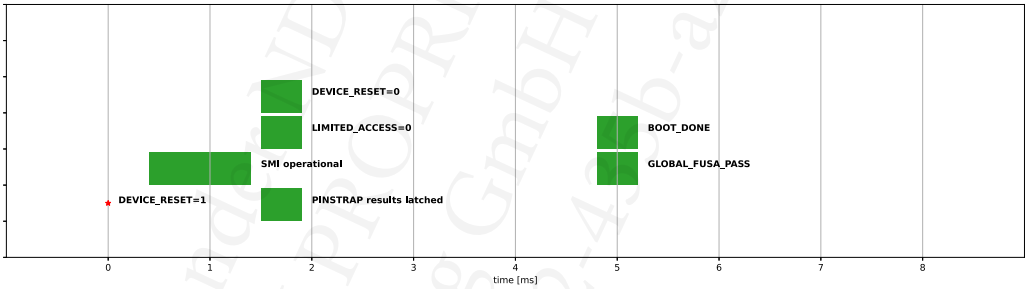


Figure 23. Startup timing: Software reset

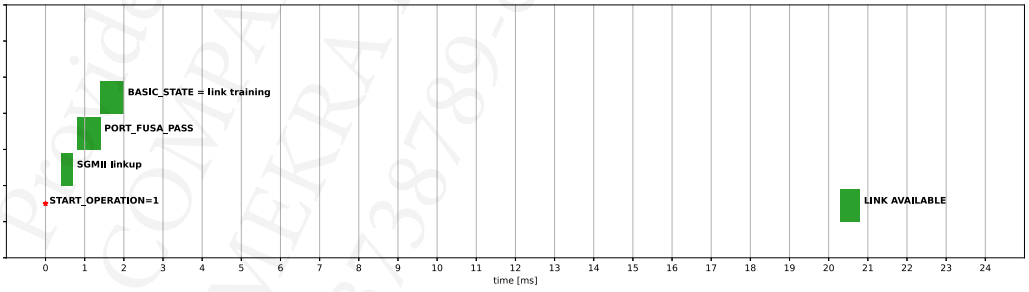


Figure 24. Start operation timing

Note: *Timings related to wake / sleep can be found in Section 5.5*

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4.2 Reconfiguration

Writing to the configuration registers is disabled by default to avoid unintended reconfiguration. To change the configuration, configuration register writing must be enabled. There are multiple, hierarchical CONFIG_ENABLE switches. Table 26 gives an overview of the registers and the way each is protected.

There is one device configuration enable bit (SUPER_CONFIG_ENABLE) that enables all reconfiguration. The configuration usually occurs after power-on as part of the software initialization routine. However, PHY reconfiguration is possible at any later time.

All other registers that are not listed in this table are not write protected.

Table 26. Reconfiguration protected registers

Level 1	Level 2	Level 3	Protected registers
MMD30.DEVICE_CONTROL.SUPER_CONFIG_ENABLE			
	MMD30.DEVICE_CONTROL.GLOBAL_CONFIG_ENABLE		
			MMD30.DEVICE_CONFIG
			MMD30.WISE_CONFIG
			MMD30.WISE_PARAMETERS
			MMD30.SMI_CONFIG
			MMD30.DEVICE_CONFIG_EXTENDED
			MMD30.SMI_DATA_IO_CONFIG
			MMD30.COMPL_TEST_CONTROL
	MMD30.GLOBAL_INFRA_CONTROL.CONFIG_ENABLE		
			MMD30.GPIO0_FUNC_CONFIG
			—
			MMD30.GPIO9_FUNC_CONFIG
			MMD30.GPIOx_IO_CONFIG
	MMD30.PORT_CONTROL.CONFIG_ENABLE		
			MMD30.PORT_FUNC_ENABLES
		Shared.PHY_CONTROL.CONFIG_ENABLE	
			MMD1.BASE_T1_PMA_CONTROL
			MMD1.E1000BT1_TEST_MODE_CONTROL
			Shared.PHY_CONFIG
			Shared.PHY_PARAMETERS
			Shared.WAKE_SLEEP_CONFIG
			Shared.WAKE_SLEEP_PARAMETERS
			Shared.COMPL_TEST_CONTROL
		MMD30.PORT_BIST_CONTROL.CONFIG_ENABLE	
			MMD30.BIST_INTERCEPT_CONFIG
			MMD30.PORT_BIST_CONFIG
		MMD30.PORT_INFRA_CONTROL.CONFIG_ENABLE	
			MMD30.INFRA_CONFIG
			MMD30.MII_BASIC_CONFIG
			MMD30.XMII_CLK_CONFIG
			MMD30.XMII_CLK_IO_CONFIG

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Table 26. Reconfiguration protected registers*...continued*

Level 1	Level 2	Level 3	Protected registers
			MMD30.XMII_DATA_CONFIG
			MMD30.XMII_DATA_IO_CONFIG
			MMD30.RGMII_RXC_DELAY_CONFIG
			MMD30.RGMII_TXC_DELAY_CONFIG
			MMD30.MONITOR_CONFIG
		MMD30.SGMII_ADVANCED_CONTROL.CONFIG_ENABLE	
			MMD30.SGMII_ADVANCED_CONFIG
		MMD30.GPTP_CONTROL.CONFIG_ENABLE	
			MMD30.PTP_CLK_NSEC
			MMD30.PTP_SYNC_PERIOD_0
			MMD30.PTP_SYNC_PERIOD_1
			MMD30.LTC_LOOP_CONTROL
		MMD30.PTP_CONTROL.CONFIG_ENABLE	
			MMD30.PTP_MESSAGE_FILTER
			MMD30.PTP_FILTER_ENABLES
			MMD30.TX_DELAY_NSEC
			MMD30.TX_DELAY_XTD
			MMD30.RX_DELAY_NSEC
			MMD30.RX_DELAY_XTD
			MMD30.USER_MAC_DA_0
			MMD30.USER_MAC_DA_1
			MMD30.USER_MAC_DA_2
			MMD30.USER_MAC_DA_MASK_0
			MMD30.USER_MAC_DA_MASK_1
			MMD30.USER_MAC_DA_MASK_2
			MMD30.FILTER_PARAMETERS

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Table 27 gives an overview of the test configuration enable hierarchy.

Table 27. Test configuration register enables

TEST_ENABLE will not affect the configuration but enables the test function.

Level 1	Level 2	Level 3	Affected test functions
MMD30.DEVICE_CONFIG.SUPER_TEST_ENABLE			
	MMD30.DEVICE_CONFIG.TEST_ENABLE		
			MMD30.COMPL_TEST_CONTROL.COMPL_TCLK_EN
	MMD30.PORT_FUNC_ENABLES.TEST_ENABLE		
		Shared.PHY_CONFIG.TEST_ENABLE	
			MMD1.PMA_CONTROL1.REMOTE_LOOPBACK
			MMD1.PMA_CONTROL1.LOCAL_LOOPBACK
			MMD1.E1000BT1_TEST_MODE_CONTROL.TEST_MODE
			MMD3.PCS_CONTROL1.LOOPBACK
			MMD3.E1000BT1_PCS_CONTROL.LOOPBACK
			Shared.TDR_CONTROL.ENABLE
			Shared.TDR_CONTROL.START
		MMD30.PORT_BIST_CONFIG.TEST_ENABLE	
			MMD30.PORT_BIST_CONFIG.PACKET_BIST_ENABLE
		MMD30.INFRA_CONFIG.TEST_ENABLE	
			MMD30.XMII_CONTROL.MAC_LOOPBACK
			MMD30.XMII_CONTROL.PHY_LOOPBACK

4.3 xMII reconfiguration

TJA1121 A supports RGMII while TJA1121B supports SGMII, hence the desired xMII basetype is chosen by selecting the respective device variant. In addition, some parameters can be configured either via pin strapping (see Table 20) or via register configuration.

On TJA1121A, the internal delay for RGMII TXC/RXC can be enabled separately and the phase shift can be configured.

On TJA1121B, the SGMII interface can be configured for PHY or MAC mode.

4.4 xMII isolation

The xMII data and control signals can be isolated, meaning disconnected between internal logic and IO-cells. There are several ways of achieving this:

- Logical isolate (RX+TX): configurable via MMD30.XMII_CONTROL.LOGICAL_ISOLATE (30.AFC0h.2) - TX data and control signals are ignored, RX data and control signals are tied LOW. Output clocks remain active.
- RXD disable (RX only): configurable via MMD30.XMII_DATA_CONFIG.RXD_DISABLE - RX data and control signals are tied LOW. Output clocks remain active.

Both isolation methods do not check for active frame transmissions. Deassertion of isolation must be done at frame booundary.

In addition to manually controlling MMD30.XMII_CONTROL.LOGICAL_ISOLATE (30.AFC0h.2), ISOLATE_ON_LINKDROP enables automatic assertion of LOGICAL_ISOLATE on every link-drop. This way accidenatal transmissions on active but possibly unconfigured links can be avoided.

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Note: *Logical isolate also inhibits data transfer from BIST module which has the following consequenes:*

- *BIST transmitter can be used to test successful assertion of logical isolate. Transmission will cease.*
- *BIST transmitter cannot be used during, or independant of, assertion of logical isolation.*

4.5 SGMII reconfiguration

The SGMII mode (PHY or MAC) can be configured via pinstrapping. Configuration via SMI allows overwriting the pinstrapping configuration as well as setting additional parameters.

Table 28 lists registers commonly used during SGMII configuration. For a full set of registers see register description in the datasheet starting MMD30 address B000h. (Section 7.4)

Table 29 lists registers shared between TJA1121A and TJA1121B which contain SGMII related fields.

Table 28. SGMII commonly used registers

Register	Description
MMD30.SGMII_BASIC_CONTROL 30.B000h	• Reset • Auto negotiation • Power down
MMD30.SGMII_BASIC_STATUS 30.B001h	• Link status • Auto negotiation status
MMD30.ADVANCED_CONTROL[CONFIG_ENABLE] 30.B010h	• Config enable • Standby/Start operation
MMD30.ADVANCED_STATUS 30.B011h	• Link status
MMD30.ADVANCED_CONFIG 30.B013h	• PHY mode (PHY or MAC) • Operation mode (autonomous or managed)
MMD30.SGMII_IRQ_SOURCES 30.B014h MMD30.SGMII_IRQ_ENABLES 30.B015h MMD30.SGMII_IRQ_MSTATS 30.B016h	Interrupt related registers • FIFO status • Autonegotiation status • Link status
MMD30.SGMII_LED_TRIGGER_0 30.B040h	SGMII LED status control
SGMII_PCS_CONFIG1 30.B073h	Polarity inversion for Tx or Rx differential pair
SGMII_PCS_STATUS 30.B072h	SGMII PCS loopback When SGMII PCS loopback is enabled, the first byte of the preamble is dropped. *Workaround*: none

Table 29. Registers containing SGMII related fields

Register	Description
Shared.PORT_FUNC_IRQ_ENABLES 30.807Ah Shared.PORT_FUNC_IRQ_STATUS 30.8078h Shared.PORT_FUNC_IRQ_MSTATUS 30.8074h	• Serdes IRQs
Shared.PORT_LEVEL_IRQ_ENABLES 30.8072h Shared.PORT_LEVEL_IRQ_STATUS 30.8070h Shared.PORT_LEVEL_IRQ_MSTATUS 30.8074h	• Serdes IRQs
CL22.FAVORITE_IRQS 30.8014h	• SGMII_LINK_EVENT

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Table 29. Registers containing SGMII related fields...*continued*

Register	Description
MMD30.LEDx_TRIG_SOURCE 30.01A1h	• Port SGMII LED Trigger
MMD30.XMII_ABILITIES 30.AFC4h	• SGMII_ABILITY
MMD30.MII_BASIC_CONFIG 30.AFC6h	• CMOS_OR_SERDES • XMII_MODE
Shared.COMPOSITE_STATUS 30.8021h	• ALL_LINK_STATUS • LINK_STATUS_SELECT

4.6 1000BASE-T1 configuration

Leader/Follower can be configured either via pin strapping or via register MMD1.BASE_T1_PMA_CONTROL.LEADER_FOLLOWER. The register is configuration protected. See Configuration protection for details. Figure 21 shows a configuration example including the configuration enable steps.

4.7 Polarity detection and correction

Polarity detection and correction is mandatory for 1000BASE-T1 and therefore enabled per default. Polarity detection is always enabled, its status is reflected by the Shared.PHY_STATUS.DETECTED_POLARITY bit. Polarity correction can be disabled manually via register setting Shared.PHY_CONFIG.POLARITY_CORRECT_DISABLE.

Note that polarity detection is performed only once when the link is established. Enabling or disabling polarity detection has no effect without reconnecting the cable or restarting the process in another way. Furthermore, Shared.PHY_CONFIG.POLARITY_SWAP does invert polarity regardless of polarity detection. This inversion has two non-obvious effects:

1. Activating polarity swap on an established link brings down the link, regardless of the polarity detection setting.
2. Activating polarity swap will invert the state of MMD1.E1000BT1_PMA_STATUS.POLARITY_DETECT / Shared.PHY_STATUS.DETECTED_POLARITY flag the next time a polarity detection is performed.

Polarity options can be configured via the Shared.PHY_CONFIG (30.8108h) register.

Table 30. PHY_CONFIG polarity relevant part

Bit	Symbol	Value	Description
3	POLARITY_SWAP	0	No polarity swap
		1	Polarity swap enabled, TRX_P/TRX_N pins swapped
2	POLARITY_CORRECT_DISABLE	0	Polarity correction enable. Only possible in Follower mode
		1	No polarity correction

Figure 25 shows how to disable the polarity detection and enable polarity swapping. Both settings can be changed individually.

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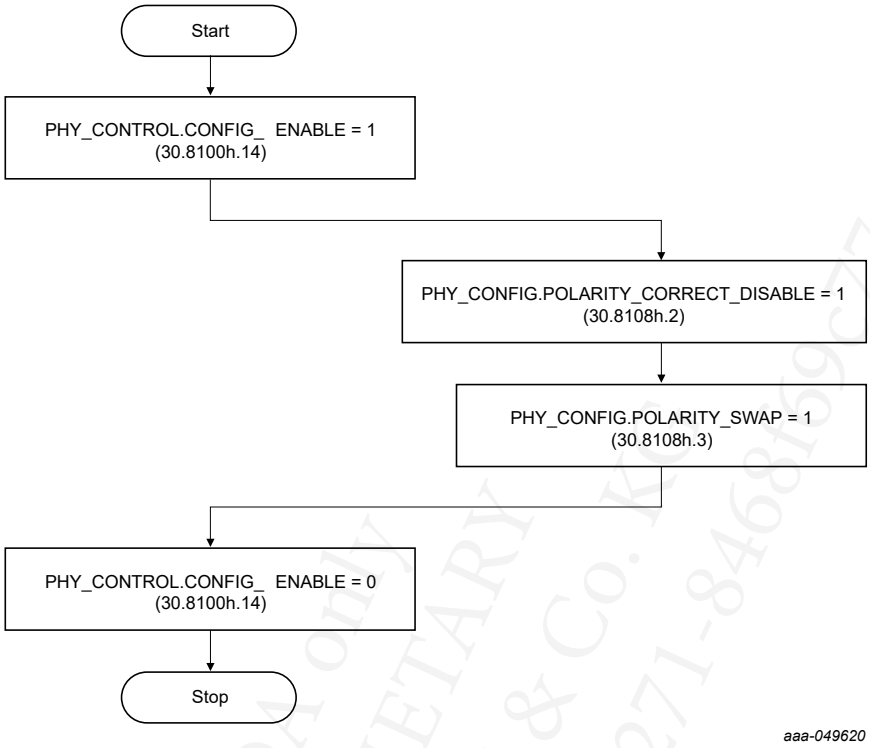


Figure 25. Example flow: Polarity settings

4.8 Frequently used registers

The following registers are worth examining during various situations.

Table 31. Registers of interest

State	Register
During link training	• MMD30.PHY_STATUS (30.8102h) – LINK_AVAILABLE (Bit 6)
On link up failure	• MMD30.PHY_STATUS (30.8102h) – SENDN_OR_DATA (Bit 5) – REM_RCVR_STAUS (Bit 4) – SCRAMBLER_STATUS (Bit 3) – LINK_STATUS (Bit 2) – LOC_RCVR_STATUS (Bit 1) • MMD30.PHY_STATE (30.810Ch) – BASIC_STATE (Bit 0:3) – TRANSMIT_STATE (Bit 8:11)
Polarity issues	• MMD30.PHY_STATUS (30.8102h) – DETECTED_POLARITY (Bit 7) • MMD30.PHY_CFG (30.8108h) – POLARITY_SWAP (Bit 3) – PLARITY_CORRECT_DISABLE (Bit 2)
SMI access issues	• CL22.ALWAYS_ACCESSIBLE (1Fh) – REFCLK_WARN (Bit 4)

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Table 31. Registers of interest*...continued*

State	Register
	– CORE_SUPPLY_STATUS (Bit 2) – LIMITED_ACCESS (Bit 1)
Diagnosis	• MMD1.PMA_CONTROL1 (01.0000h) – REMOTE_LOOPBACK (Bit 1) – LOCAL_LOOPBACK (Bit 0) • MMD3.PCS_CONTROL1 (03.0000h) – LOOPBACK (Bit 14) • MMD30.XMII_CONTROL (30.AFC0h) – MAC_LOOPBACK (Bit 1) – PHY_LOOPBACK (Bit 0) • MMD1.E1000BT1_TEST_MODE_CONTROL (01.0904h) – TEST_MODE (Bit 15:13)
Leader/Follower	• MMD1.BASE_T1_PMA_CONTROL (01.0834h) – Leader_Follower (Bit 14)

4.9 Operating Modes

The following table provides information about the status of all pins during various operating modes. Transitions between modes are highlighted in the data sheet under Section 6.1, *Operating modes*.

Table 32. Operating Modes

Pin	Type	Power off	Deep sleep	Standby Sleep Disable	Operational
Requirements					
VDD_CORE VREGA_IN VDDA_RX VDDA_TRX VDDIO		Off	Off	On	On
VDDA_AO		Off	On	On	On
Clock		Off	Off	On	On
RST_N		Don't care	High	High	High
Pin states					
MDC MDIO	Input/Output[1]	Unpowered	Unpowered	Active	Active
INT_N	Output[1] Low side open drain	Unpowered	Unpowered	Active	Active
XI XO		Unpowered	Unpowered	Active [2]	Active [2]
RST_N	Input[3] Internal pullup	Unpowered	Active	Active	Active
WAKE_IN_OUT	Input/Output[3]	Unpowered	Floating[4]	Floating[4]	Floating[4]

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Table 32. Operating Modes*...continued*

Pin	Type	Power off	Deep sleep	Standby Sleep Disable	Operational
	High side open drain				
INH	Output[3] High side open drain	Unpowered	Floating	High[5]	High[6]
TRX_P TRX_M	Analog[7]	Unpowered, high impedance	No bias, 100Ω differential termination	No bias, 100ohm differential termination	100ohm differential termination 0.9 V bias
GPIO0	Input/Output[1]	Unpowered	Unpowered	Output[4]	Output[4]
GPIO1 GPIO2 GPIO9	Input/Output[1]	Unpowered	Unpowered	Floating[4]	Floating[4]
RXD0-RXD3 RX_CTL RXC	Output[1]	Unpowered	Unpowered	Output, 50ohm	Output, 50ohm
TXD0-TXD3 TX_CTL TXC	Input[1]	Unpowered	Unpowered	Input, high impedance	Input, high impedance

[1] Referenced to VDDIO, ESD diodes to GND and VDDIO present
[2] Depending on configuration either single ended input, differential input or crystal oscillator circuit
[3] Referenced to VDDA_AO, ESD diodes to GND and VDDA_AO present
[4] Power-on behavior can be changed via configuration.
[5] During reset, INH is asserted when RST_N is asserted..
[6] During reset, INH is asserted when RST_N is asserted.
[7] Referenced to VDDA_TRX, ESD diodes to GND and VDDA_TRX present

5 Extended Control

5.1 Test modes

PMA test modes supported by this device can be enabled via MMD1.E1000BT1_TEST_MODE_CONTROL.TEST_MODE. The following test modes can be selected:

Table 33. Register differences

Value	Name	Description	Purpose
0	Normal		No test mode activated
1	Test mode 1	Provide 125 MHz test clock on TX_TCLK pin	Transmit clock jitter test.
2	Test mode 2	Transmit a continuous pattern of 3x {+1} followed by 3x {-1} which results in a 125 MHz signal	Transmit MDI jitter test
3	Reserved		
4	Test mode 4	Transmit symbol sequence defined by $g(x) = 1 + x^9 + x^{11}$ (5)	Transmit distortion test
5	Test mode 5	Transmit Interframe idle signals	PSD mask test

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Table 33. Register differences...*continued*

Value	Name	Description	Purpose
6	Test mode 6	Transmit a continuous pattern of 15x {+1} followed by 15x {-1} which results in a 25 MHz signal	Transmitter droop test
7	Test mode 7	Perform linkup and transmit zero data pattern instead of xMII data. Expect zero data pattern on receiver and count bit errors after FEC + 80B81B decoding.	Bit error rate measurement / RS-FEC encoder/decoder test

```
//Recommended: reset PHY
30.0040h = 8000h
//Wait at least 10ms or check DEV_BOOT_DONE
//Set Superconfig to allow configuration changes
30.0040h = 2000h
//Set Supertest to allow test modes
30.0048h = 8002h
//Set testmode (test mode number left-shifted by 13
//in this example test mode 6)
01.0904h = C000h
//Start operation
30.0040h = 2001h
```

5.1.1 Recovered clock (TX_TCLK)

Some compliance test require access to the TX_TCLK. In slave linked state this is obviously not possible via MDI so a dedicated pin can be configured to output the recovered TX_TCLK. For both TJA1121A and TJA1121B it is provided via pin 26. On the A-variant it is multiplexed with RXC so it can't be accessed while RGMII is activated. On the B-variant it is multiplexed with GPIO8.

The following code sequence shows how to enable TX_TCLK. It can be combined with other test modes.

```
//Recommended: reset PHY
30.0040h = 8000h
//Wait at least 10ms or check DEV_BOOT_DONE
//Set Superconfig to allow configuration changes
30.0040h = 2000h
//Set Supertest to allow test modes
30.0048h = 8002h
//Set COMPL_TCK_EN in MMD30.COMPL_TEST_CONTROL
30.03F0h = 0001h
//Start operation
30.0040h = 2001h
```

5.2 Loopback

Loopback operation can be used for diagnosis purposes, especially to locate potential error sources. Ethernet frames are transmitted by the MAC and looped back at some point of the data path. If the loop is error-free, the MAC receives the same frames as sent out before.

In accordance with the 1000BASE-T1 IEEE specifications, the TJA1121 provides the following loopback modes: Internal, external, xMII, and remote loopback.

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In case of internal loopback, the loop point is after the PCS block. It checks proper operation of the internal PCS part. In case of external loopback, the loop point is after the PMA block. It checks proper operation of both the internal PCS and PMA part. The remote loopback spans the entire loop from one PHY to the other PHY and back to the first PHY. It checks the entire channel.

Table 34 gives information about activating different loopback modes.

Table 34. Loopback modes

Mode	Register	Precondition	Additional Info
Internal (PCS)	MMD3.PCS_CONTROL1. LOOPBACK (03.0000h.14)		Link available not set when activated
External (PMA)	MMD1.PMA_CONTROL1. LOCAL_LOOPBACK (01. 0000h.0)	• cable must be open, not terminated or connected to link partner	Link available set when cable is open
Remote (PMA)	MMD1.PMA_CONTROL1. REMOTE_LOOPBACK (01. 0000h.1)	• Link partner connected	Link available set when link partner connected
xMII PHY	MMD30.XMII_CONTROL. PHY_LOOPBACK (30.AFC0h. 0)	• INFRA_CONFIG. TEST_ ENABLE set • Link partner connected	Link available set when link partner connected
xMII MAC	MMD30.XMII_CONTROL. MAC_LOOPBACK (30.AFC0h. 1)	• INFRA_CONFIG. TEST_ ENABLE set	Link available set when link partner connected

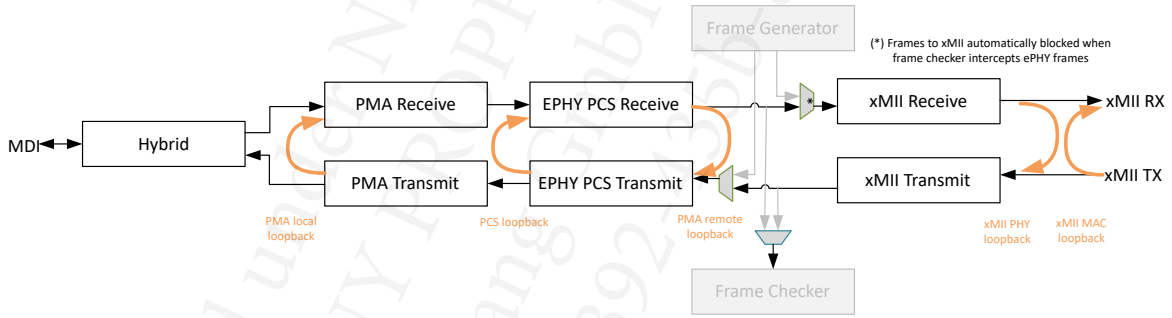


Figure 26. Supported loopbacks

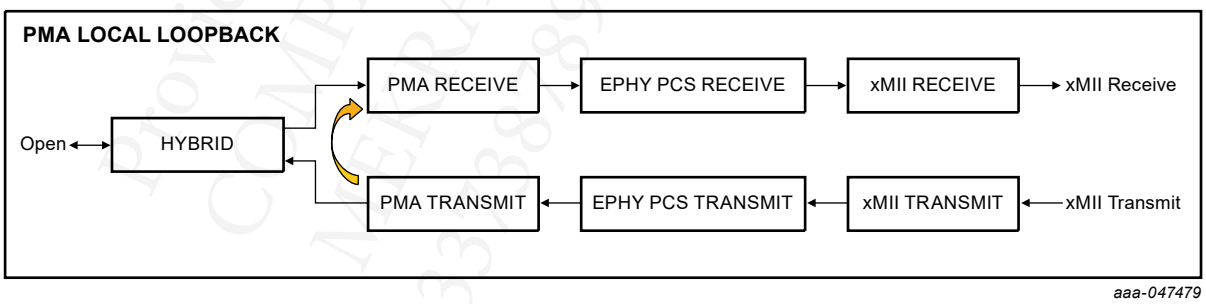


Figure 27. External (PMA local) loopback

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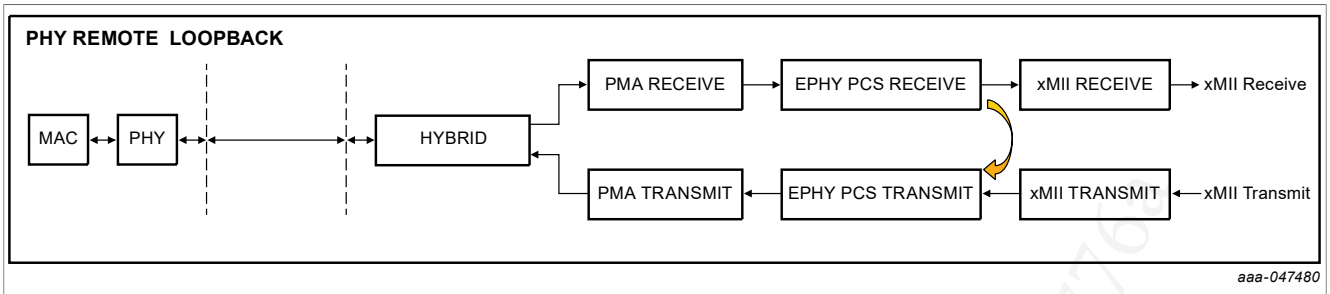


Figure 28. Remote (PMA remote) loopback

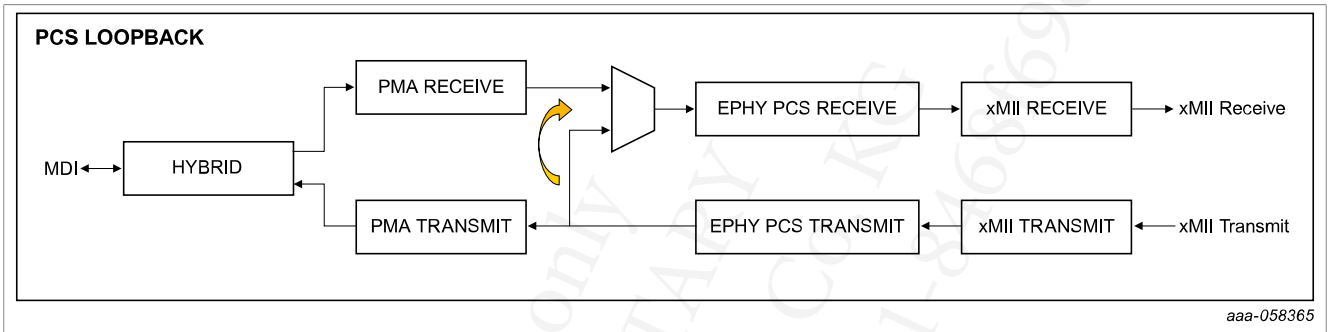


Figure 29. Internal (PCS) loopback

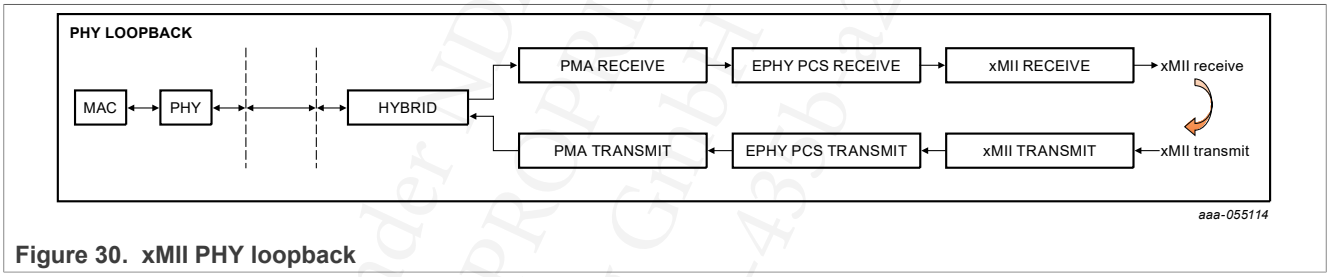


Figure 30. xMII PHY loopback

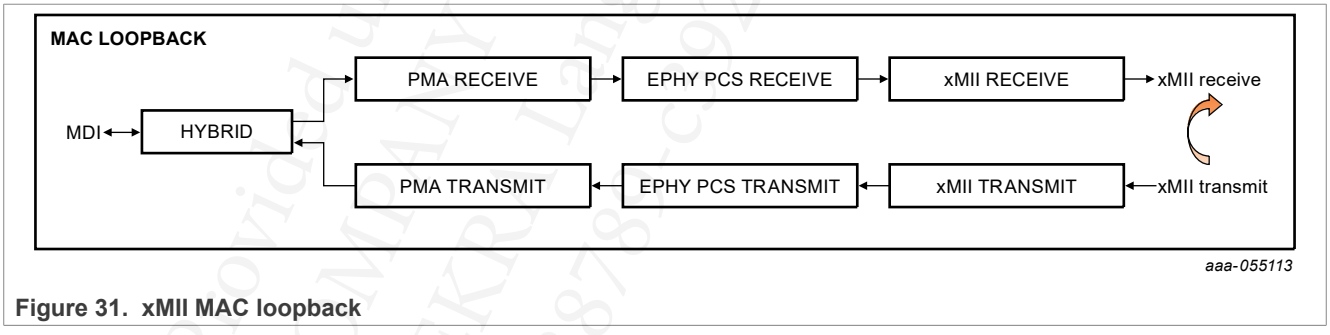


Figure 31. xMII MAC loopback

5.3 Timestamping

The TJA1121 contains a IEEE 1588 v2 (PTP) module which allows timestamping inbound and outbound frames with high precision. The following diagram shows the main function blocks participating in the ptp operation.

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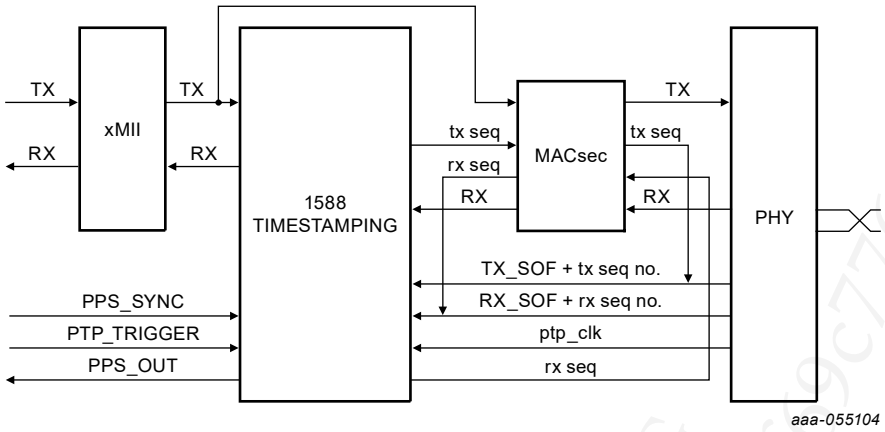


Figure 32. PTP function blocks

To get high precision timestamps the internal local time counter (LTC) must either be rate corrected to an externally provided reference clock, pose as the reference clock itself or ensure synchronized operation by other means.

- The rate correction can be achieved by 2 methods:
  - Hardware self-locking clock (1pps PLL)
  - Software assisted rate correction
- Using the LTC as reference for external devices can be achieved by using the PPS_OUT signal.

5.3.1 Hardware self-locking clock

Using the hardware self-locking clock requires the PPS_SYNC to be connected to a reference clock with known frequency and known absolute timestamps for every edge of the signal. The operation of the hardware self-locking clock works in three steps:

1. Basic initialization, for example, enabling PTP and setting PTP_CLOCK_PERIOD
2. Initializing the LTC with a correct timestamp
3. locking the LTC to a reference signal

The initialization of the LTC works by first setting the interval of the PPS_SYNC signal in the HW_EXT_PPS_NS_x registers (0 s - 1.073 s in 1 ns steps). The duty cycle does not matter as only the rising edge is used for synchronization. As a safeguard, the register HW_LOCK_ER_LMT_x can be used to set the maximum allowed deviation between the external PPS_SYNC and the internal nominal LTC. (0 ms - 16 ms in 1 ns steps) An interrupt is asserted when the limit is exceeded.

The next step is to load the timestamp of any upcoming rising edges of the PPS_SYNC signal into the LTC_WR_NSEC_x / LTC_WR_SEC_x registers. Finally, the LOAD_LTC_PPS bit in the LTC_LOAD_CTRL register must be set at an arbitrary time in the clock cycle preceeding the chosen rising edge. The timestamp is loaded into the LTC when the next rising edge of the PPS_SYNC signal occurs.

The locking of the LTC is done automatically.

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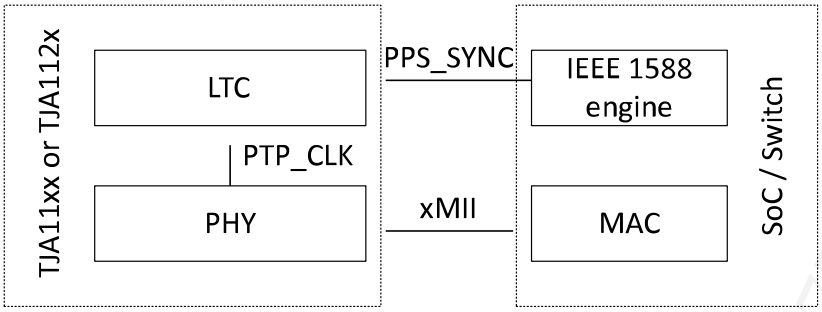


Figure 33. PTP hardware signals

5.3.2 Software assisted rate correction

Alternatively to the hardware self-locking clock a software-assisted rate correction can be used. To match rate differences as well as correct offsets a constant rate correction and a temporary rate correction is required. The details of those concepts are described in the data sheet [1] under section 6.11.5.2 / 6.11.5.3.

To calculate these rate correction values, the ltc value and reference timestamp of some event must be known. This calculation can be achieved by using the event timestamping mechanism, which requires connecting the PTP_TRIGGER signal to the host controller.

5.4 Time domain reflectometry

In order to be able to detect errors in the ethernet cable and calculate the approximate error location the PHY implements TDR (time domain reflectometry) as diagnostic feature. TDR works by sending a pulse on the differential signal pair and observes the received signal. By analyzing the timing and polarity of the received signal the cable can be characterized.

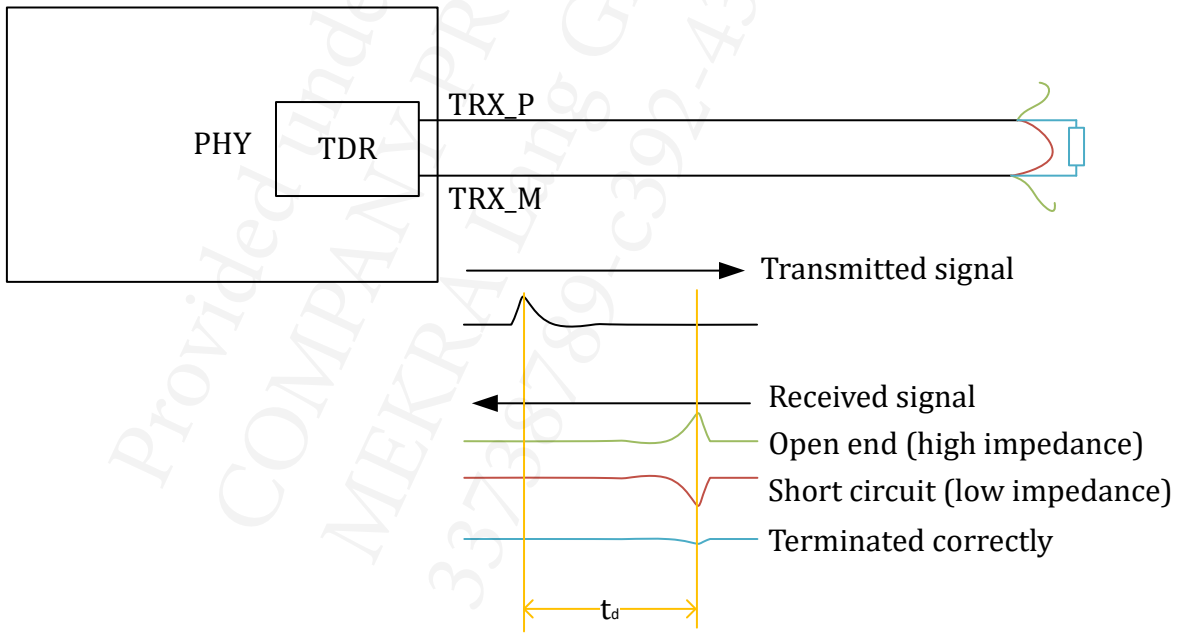


Figure 34. Time domain reflectometry concept

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The implementation in the PHY is able to detect 4 different states:

- No fault: This usually means the cable is terminated properly on the far end. This test cannot give information about whether a link partner is connected and working properly
- Open cable: The test identified a signature similar to an open cable. Furthermore the test is able to measure the approximate length of the open cable
- Shorted cable: The test identifiied a signature similar to a short between TRX_P and TRX_N. Furthermore the test is able to measure the approximate location of the short.
- Unable to detect: This indicates that no measurement was possible. Most likely due to a link partner that is transmitting.

5.4.1 TDR operation

The following sequence is required to use TDR:

```
//Allow change of some register
DEVICE_CONTROL.SUPER_CONFIG_ENABLE (30.0040h.13) = 1

//Enable test functions
EPHY_CONFIG.TEST_ENABLE (30.8108h.1) = 1

//Change TDR pulse frequency
WAVELET_CONFIG.LOG2_T1 (1.8367h.5:3) = 1

//Enable and start TDR, can be done in one register write
TDR_CONTROL.INCLUDE_LINK_TEST (30.8360h.12) = 1
TDR_CONTROL.ENABLE  (30.8360h.15) = 1
TDR_CONTROL.START  (30.8360h.13) = 1

//Start PHY (this triggers the actual TDR measurement)
DEVICE_CONTROL.START_OPERATION (30.0040h.0) = 1

//Read TDR_STATUS until VALID flag is true
TDR_STATUS.VALID (30.8361h.15) == 1

//Read state+fault type
TDR_STATUS.TDR_STATE (30.8361h.14:12)
TDR_STATUS.FAULT_TYPE (30.8361h.2:0)
```

Note: *Execution time of TDR is approximately 1-1.5 seconds*

5.5 Wake / Sleep (TC10)

The TJA1121 allows 2 different Wake-sleep concepts which can be configured via Shared.WAKE_SLEEP_CONFIG.FUNCTION_SELECT (30.8182h.15:14):

- **Basic Wake-Sleep**
  In this mode the device can be put into sleep state as well as into low power sleep, however there is no sleep handshake performed on the ethernet link. In case there is a link established when the device is put into sleep the link breaks up without further notice/interaction. Wake-up is only possible via local signals (WAKE_IN_OUT pin or WU_SYS/WU_PHY via SMI).
- **TC10 Wake-Sleep**

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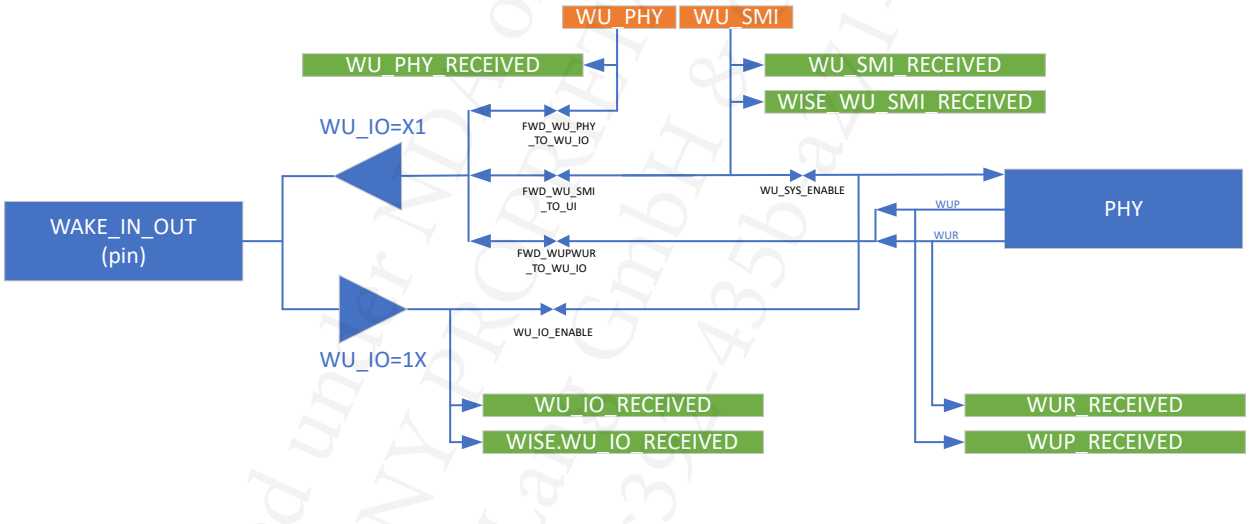
In this mode Wake-Sleep according to TC10 standard is implemented. In addition to the functions described under *Basic Wake-Sleep* the following functions are implemented:

- On sleep requests the device perfoms a handshake with the link partner requesting it to also go into sleep. In case the handshake fails or there is no link established the device will not go into sleep.
- The device can receive a sleep request from the link partner via ethernet. It can be configured to automatically accept or reject the request after a certain time. Manual accepting or rejecting before the time limit is reached is also possible.
- The device can be woken up via ethernet by a wakeup pulse (WUP). In addition the WUP can be forwarded to the WAKE_IN_OUT pin.
- The device can send a WUP on a sleeping link.
- The device can transmit a wakeup request (WUR) on an established link

TC10 Manual sleep: In *TC10 Wake-Sleep* a sleep request automatically triggers the described sleep request. In order to perform a manual sleep without initiating a handshake the device can be manually put into standby. A subsequent sleep request will directly put the device into sleep state.

TC10 Sleep without link: In *TC10 Wake-Sleep* a sleep request is only executed when a link is present. In case a sleep request is issued before a link is present it is *stored* indefinetely and only executed once the link is up.

The following diagram shows configurable wake trigger interconnects.



Orange: Trigger bit in register

Green: Status bits

Black: Enable bit in register

Figure 35. TC10 wakeup interconnects

5.5.1 Timers

There are several timers related to Wake / Sleep.

Table 35. TC10 timers

Name	Type	Duration	Description
SLEEP_ACK_ TIMER	Configurable	0us - 12ms	Defines the time from reception of LPS until sleep is automatically accepted or rejected. In case of accept a LPS is sent (See TC10 FSM in datasheet)

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Table 35. TC10 timers...*continued*

Name	Type	Duration	Description
AUTO_SLEEP_ IDLE_TIMER	Configurable	125ms - 16s	Defines the time a link must be idle before a sleep request is automatically sent
AUTO_SLEEP_ SILENCE_TIMER	Configurable	125ms - 16s	Defines the time a link must be silent after wake-up before the PHY returns to sleep. (This shall catch fals remote wakeup conditions)
ACK_TIMER	Specification	8ms	This value is the expected time a PHY responds with a LPS response to a sleep (LPS) request.
REQ_TIMER	Internal	16ms	This timer is started when a sleep request is issued and an LPS is sent. when an LPS is received and the link becomes silent the sleep is successful. If this timer expires before an LPS is received and the link becomes silent the sleep did not succeed and a sleep fail is indicated.
WUP_TIMER	Specification	0.5ms - 1.5ms	A timer used for teliable detection of WUP pulses.
TWU_LINK_ passive	Specification	<2ms	This is the maximum time it may take to send a WUP over a passive link
TWU_LINK_active	Specification	<1ms / <3ms[1]	This is the maximum time it may take to send a WUR over an active link

[1] In case of jumbo packets (up to 16kB) 3ms are allowed instead of 1ms

5.6 MACsec

The TJA1121A/B includes a MACsec block. This block implements Ethernet frame encryption and authentication as described in the IEEE802.1AE standard.

The MACsec functionality is split into two sub modules:

1. The *MACsec Adapter*, which connects the RX and TX data path to the MACsec IP
2. The *MACsec IP*, which implements the actual MACsec functionality

Both MACsec sub modules can be configured using the SMI interface and are mapped into the MMD31 region. However, all MACsec registers are 32-bit wide, while SMI only offers 16-bit wide access. The next section introduces the register access to the MACsec block.

5.6.1 Register access

All MACsec registers are mapped into MMD region 31 as shown in Table 36. The MACsec registers can be found starting from SMI address 9000h up to address 9FFFh. The registers for the MACsec adapter and the MACsec IP can be found inside this region. Three points must be considered when accessing these registers:

1. All MACsec registers as they are described in the data sheet are 32bit wide. However, SMI only allows 16bit access.
2. All MACsec register addresses in the data sheet are given as byte address. This results in an increment (of the byte address) of four between two neighboring MACsec registers. Accordingly, the SMI address difference between two neighboring MACsec registers is two.
3. All MACsec register addresses are given as relative byte address offset to the base address of each sub module (MACsec adapter or MACsec IP). See table Table 37.

Table 36 shows how the different address spaces are mapped into each other. Additionally, it also shows how to access a complete 32bit MACsec register, with two consecutive SMI accesses. The lower half (Bit 0…15) of each 32bit MACsec register can be accessed with an even SMI address, while the upper half (Bit 16…31) is accessed with an odd SMI address.

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Table 36. 32-bit MACsec registers in MMD31

MACsec register (32bit word)			SMI Address (16bit)	MACsec register byte address
reserved				
MACsec	Word 2047	Upper	9FFFh	1FFCh
		Lower	9FFEh	
	⋮			
	Word N	Upper	9001h + N × 2	N × 4h
		Lower	9000h + N × 2	
	⋮			
	Word 1	Upper	9003h	0004h
		Lower	9002h	
	Word 0	Upper	9001h	0000h
		Lower	9000h	
reserved				

The registers addresses for each MACsec sub module are given in the data sheet as byte offsets relative to each MACsec sub module. Table 37 shows the offset for each MACsec sub module.

Table 37. Address offset of MACsec sub modules

Name	Relative sub module offset (in bytes)	Absolute SMI offset
MACsec adapter	0F00h	9780h
MACsec IP	0000h	9000h

The SMI address of the lower half word can now be calculated using the MACsec sub module offset (in bytes) and the register offset inside the MACsec sub module (in bytes):

$$\text{SMI address} = 9000\text{h} + \frac{\text{sub module offset} + \text{register offset}}{2}$$

The upper-half word can be accessed at the subsequent SMI address.

Example: The ADAPTER_STATUS register in the MACsec adapter is at offset 004h. The resulting SMI address of the lower half word is:

$$9000\text{h} + \frac{0\text{F00h} + 004\text{h}}{2} = 9782\text{h}$$

Consequently, the address of the upper-half word is 9783h.

The following pseudo code shows a function to read a 32bit MACsec register:

```
function macsec_read(sub_module_offset, register_offset):
    smi_address_lower = 9000h + (sub_module_offset, register_offset)/2

    lower = smi_read(MMD31, smi_address_lower)
    upper = smi_read(MMD31, smi_address_lower + 1)

    return (upper << 16) + lower
```

Note that for reading registers a SMI access with address autoincrement can also be used.

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5.6.2 Example configuration

This example demonstrates how to configure the MACsec subsystem to send confidentiality and or integrity protected Ethernet frames. For this, both the MACsec adapter and the MACsec IP must be configured. The MACsec adapter is enabled as the following:

- Set ADPTR_ENA bit in ADPTR_CNTRL

The MACsec IP is configured next:

- Select a Secure Channel of the MACsec IP TX_SC_ID = 0…31
  - ***Note:*** *This ID is used to select the internal registers of the MACsec IP, it is not the Secure Channel Identifier used in the SecTAG of the Ethernet frame*
- Enable access to the SAK and SAH registers by writing the TX_SC_ID to TX SC access select (MACSEC_TXSCKA)
  - Write a key in the Security Association A via:
    - TX Security Association A key 1 of 8 (MACSEC_TXSAKA1Q)
    - TX Security Association A key 2 of 8 (MACSEC_TXSAKA2Q)
    - and so on
- Enable access to the Secure Channel configuration registers by writing the TX_SC_ID to TX SC access select (MACSEC_TXSCA)
  - Initialize the TX Security Association A nextPN (MACSEC_TXSAANPN) to 1
  - Initialize the TX Security Association Extended A nextPN (MACSEC_TXSAAXNPN) to 0
  - Activate the newly configured Security Association by clearing the A bit in TX Security Association A configuration and status (MACSEC_TXSAACS)
  - Set the Secure Channel Identifier to be used in the SecTAG
    - TX Secure Channel Identifier first half (MACSEC_TXSCI1H)
    - TX Secure Channel Identifier second half (MACSEC_TXSCI2H)
  - Configure the SC TX Secure Channel configuration (MACSEC_TXSCCFG):
    - Select the used cipher with CS
    - Configure CE to for confidentiality protection
    - Set AIS to always send the SCI
    - Configure (VLAN)-Tag handling with TCM and TVM as needed
    - Set PF to enable protection
- Configure the TX Secure Channel Selection, which classifies TX frames to a Secure Channel, for example based on the source MAC address:
  - Write the source MAC address (i.e., the address of the MAC the TJA1121 is connected to) to MAC_SA in TX SC Selection Record 0 part 0-3 (MACSEC_TXSCSR0Px)
  - In TX SC Selection Record 0 part 3 (MACSEC_TXSCSR0P3)
    - Set USA to enable source MAC address matching
    - Set EN to enable this record
    - Write your TX_SC_ID into the TX_SC field
  - With this configuration, all matching frames are now selected to be sent using the previously configured Secure Channel
- Disable Bypass Mode by setting BYPN in MACsec configuration (MACSEC_CFG)

The MACsec adapter and MACsec IP are now configured to transmit frames.

5.7 Temperature warning limit

The device monitors contain a temperature warning limit. When the predefined threshold is exceeded, MMD30.TEMP_STATUS.HIGH_TEMP_NOW is asserted and an interrupt is generated. There is no

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automatic shutdown triggered. The host controller needs to monitor the respective bit and disable the PHY on overtemperature to protect the PHY from damage.

The temperature warning threshold can be adjusted via MMD30.TEMP_STATUS.TEMP_LIMIT. The formula for calculating the register value corresponding to a given temperature is:

```
T = 7.22*(TEMP_LIMIT/100)^2 + 38.8*(TEMP_LIMIT/100) + 65.3
```

where T is the temperature in degrees Celsius and TEMP_LIMIT is the value specified in the register. Due to the temperature range supported by the sensor only register values between 41h and BFh (93°C - 165°C) correspond to valid limit temperatures.

The following linear approximation results in less than 1.5°C error between 93°C and 165°C.

```
T = TEMP_LIMIT*1.75 - 96.5
```

The preconfigured register value is A1h which corresponds to 146°C. Table 38 lists relevant register/ temperature pairs.

Table 38. Temperature limit values

Register	Temperature	Comment
0h - 40h	65°C - 93°C	Temperature range not covered by the sensor
41h	93.5°C	Lowest temperature selectable
4Fh	100°C	
A1h	146°C	Predefined default
A7h	150°C	
BFh	165.7°C	Highest temperature selectable
C0h - FFh	166°C - 211°C	Temperature range not covered by the sensor

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6 Compliance Tests

Several tests performed by external test houses guarantee interoperability between different vendors as well as compliance with various EMC guidelines. The following sections describe which settings influence the behavior of the TJA1121 towards these compliance tests.

6.1 PMA tests

The following table lists a possible set of configurations to perform all PMA tests. If some configurations are combined as mentioned in the notes a mimumum of 5 hardware configurations need to be set up in order to perform all PMA tests. The proposed set in just an example though, as several tests can be executed in different setups. (E.g. MDI passive 100 ohm termination with attached high impedance probe or MDI to 2x 50ohm probe which results in implicit 100 ohm differential termination) Also it might be desirable not to combine setups to achieve maximum precision.

Table 39. PMA test setups

Configuration	Applicable Test	Comment
Group 1: Electrical measurements		
Setup 1 MDI to DSO. MDI 100 ohm termination. DSO high impedance probe ***Note:*** *Tests can be performed on Setup 2 when disturber can be disabled and provide reliable 100ohm termination*	97.1.1 Maximum output droop	DUT must be configured to test mode 6
	97.1.2 Transmitter distortion without disturber	DUT must be configured to test mode 4
	97.1.3 Trasmitter timing jitter case 3: MDI	DUT must be configured to test mode 2
	97.1.4 Transmitter power spectral density	DUT must be configured to test mode 5
	97.1.5 Transmitter peak differential output	DUT must be configured to test mode 5
	97.1.6 Transmit clock frequency	DUT must be configured to test mode 2
Setup 2 MDI to differential voltage generator + high impedance differential probe TX_TCLK125 of DUT connected to differential voltage generator and DSO	97.1.2 Transmitter distortion with disturber	DUT must be configured to test mode 4 TX_TCLK output must be enabled
Setup 3 MDI connected to link partner (duplicate of DUT board) TX_TCLK125 of DUT connectected to DSO ***Note:*** *Tests can be performed on Setup 6 when external clock is set to nominal frequency (25 MHz)*	97.1.3 Trasmitter timing jitter case 1: MASTER	DUT must be configured to test mode 1 TX_TCLK output must be enabled
	97.1.3 Trasmitter timing jitter case 1: SLAVE	DUT must be configured to test mode 1 TX_TCLK output must be enabled
Group 2: PMA receive tests		
Setup 4 Same as **Setup 3** with different cables (A1, A2, A3). TX_TCLK125 output not required	97.2.1 Bit error rate verification	Recommended to use loopback in DUT
Setup 5	97.2.2 Alien crosstalk noise rejection	Recommended to use loopback in DUT

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Table 39. PMA test setups...*continued*

Configuration	Applicable Test	Comment
Same as **Setup 4** with noise injector added to cable		
Setup 6 Same as **Setup 4** with external clock connected to Link partner	97.2.3 Receiver frequency tolerance	Recommended to use loopback in DUT
Group 3: MDI impedance requirements		
Setup 7 MDI to VNA, 2 port (TRX_P to single ended, TRX_M to single ended)	97.3.1 MDI return loss	DUT must be configured as slave VNA output power must be below -5dBm[1]
	97.3.2 MDI mode conversion loss	DUT must be configured as slave VNA output power must be below -5dBm[1]

[1] Higher VNA power levels might trigger the PHY's sync pulse detection causing the PHY to transmit which will interfere with measurement results

7 PIN_FMEA

Table 40. TJA1121A

Failure Classification
A: *Damage to device / Bus may be affected*
B: *No damage to device / No bus communication possible*
C: *No damage to device / Bus communication possible / Corrupted node excluded from communication*
D: *No damage to device / Bus communication possible / Reduced functionality of device*
— not applicable

Pin	Pin name	Clas	Pin	Pin name	Mode	Clas	Mode	Clas	Mode	Clas	Mode	Clas	Mode	Clas	Mode
1	MDC	D	2	INT_N	Loss of IRQ and MDIO functionality	D	Loss of IRQ and MDIO functionality	D	Loss of IRQ and MDIO functionality	D	Loss of IRQ and MDIO functionality	D	Loss of IRQ and MDIO functionality	D	Loss of IRQ and MDIO functionality
2	INT_N	A	3	VDD_ CORE	Limiting value of VDD_CORE exceeded	D	Loss of IRQ functionality. IRQ constantly asserted	D	Device in undervoltage	D	Loss of IRQ functionality. IRQ constantly deasserted	D	Loss of IRQ functionality	D	Loss of IRQ functionality. IRQ constantly deasserted. Potential damage to SoC depending on limiting values on IRQ pin.
3	VDD_ CORE	D	4	VDD_ CORE	Limiting value of VDD_CORE exceeded	B	Device in undervoltage	B	Device in undervoltage	A	Limiting value of VDD_CORE exceeded	—		A	Limiting value of VDD_CORE exceeded
4	VDD_ CORE	A	5	XO	Limiting value of XO exceeded	B	Device in undervoltage	B	Device in undervoltage	B	Device in undervoltage if VDDIO is not sufficiently high. Potential damage of IO pins due to increased VDDIO	B	Device in undervoltage	D	Reduced EMC / supply filtering performance
5	XO	B	6	XI / CLK_IN	No digital clock present	B	No digital clock available, no communication possible	B	No digital clock available, no communication possible	A	Limiting value of XO exceeded	B	No digital clock available, no communication possible	A	Limiting value of XO exceeded
6	XI / CLK_IN	A	7	VREGA_ IN	Limiting value of XI exceeded	B	No digital clock available, no communication possible	B	No digital clock available, no communication possible	A	Limiting value of XO exceeded	B	No digital clock available, no communication possible	A	Limiting value of XO exceeded
7	VREGA_ IN	A	8	VDDA_RX	Limiting value of analog transmitter supply exceeded	B	No communication possible	B	No communication possible	B	If VDDIO is lower than 3.3V, no communication possible	B	No communication possible	D	Reduced EMC / supply filtering performance
8	VDDA_RX	A	9	VDDA_AO	Limiting value of analog transmitter supply exceeded	B	AFE not powered. No communication possible	B	AFE not powered. No communication possible	A	Limiting value of analog transmitter supply exceeded if VDDIO larget than 1.8V	B	AFE not powered. No communication possible	A	Limiting value of analog transmitter supply exceeded
9	VDDA_AO	D	10	RST_N	Loss of RST_N control via SoC. Internal POR	B	Always on domain unpowered. No	B	Always on domain unpowered. No	B	Depends on level and state of VDDIO. For VDDIO lower than	B	Always on domain is in undervoltage.	—	

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Table 40. TJA1121A*...continued*

Failure Classification

***A:** Damage to device / Bus may be affected*

***B:** No damage to device / No bus communication possible*

***C:** No damage to device / Bus communication possible / Corrupted node excluded from communication*

***D:** No damage to device / Bus communication possible / Reduced functionality of device*

— not applicable

Pin	Pin name	Class	Pin	Pin name	Mode	Class	Mode	Class	Mode	Class	Mode	Class	Mode	Class	Mode
					and SW remains functional		communication possible		communication possible		3.3V, the always on domain is in undervoltage.				
10	RST_N	B	11	WAKE_ IN_OUT	Device will be kept in reset due to WAKE_IN_ OUT pulldown	B	Device is reset	D	Loss of RST_N control via SoC. Internal POR and SW remains functional	B	Depends on level and state of VDDIO. Device will be kept in reset for sufficiently low VDDIO levels.	B	Spurious device reset	D	Loss of RST_N pin function. SW reset still operational
11	WAKE_ IN_OUT	B	12	INH	If INH is used and connected to SBC/SMPS, potential loss of voltage supply. No effect, if INH and wake/sleep is not used in application	D	If wake/sleep not used, no effect. Loss of wake_in_ out mechanism	D	If wake/sleep not used, no effect. Loss of wake_in_ out mechanism and spurios wakeups if internal pull-down is not enabled	D	If wake/sleep not used, no effect. Possible corruption of SoC IO pins (VDDIO domain) via INH (always-on domain) Spurious wake/ wake_forwarding events.	D	If wake/sleep not used, no effect. Spurious wake/ wake_forwarding events.	D	If wake/sleep not used, no effect. Spurious wake/ wake_forwarding events.
12	INH	A	13	VDDA_ TRX	Limiting value of analog transmitter supply exceeded	D	No effect, if wake/ sleep not used. Early SMPS/SBC shutdown and loss of supply	D	No effect, if wake/ sleep not used. Early SMPS/SBC shutdown and loss of supply	D	No effect, if wake/ sleep not used. Potential loss of voltage supply if INH is connected to SBC/SMPS	D	No effect, if wake/ sleep not used. Potential loss of voltage supply if INH is connected to SBC/SMPS	D	No effect if wake/ sleep not used. Loss of INH shut-down functionality.
13	VDDA_ TRX	B	14	TRX_P	Loss of communication	A	Loss of communication. Device may get damaged	B	No communication	A	No communication. Limiting value of analog transmitter may get exceeded	B	No communication	A	No communication. Limiting value of analog transmitter may get exceeded
14	TRX_P	B	15	TRX_M	Loss of communication	B	Loss of communication	B	Loss of communication	B	Loss of communication	B	Loss of communication	B	Loss of communication
15	TRX_M	B	16	VDDA_ TRX	Loss of communication	B	Loss of communication	B	Loss of communication	B	Loss of communication	B	Loss of communication	B	Loss of communication
16	VDDA_ TRX	B	17	GPIO0 / TMS / CONFIG0	Loss of communication	A	Loss of communication. Device may get damaged	B	No communication	A	No communication. Limiting value of analog transmitter may get exceeded	B	No communication	A	No communication. Limiting value of analog transmitter may get exceeded
17	GPIO0 / TMS / CONFIG0	D	18	GPIO1 / TDI	Loss of GPIO0/1 function, Loss of JTAG function, Wrong PHY address latching. No MDIO access / potentially conflicting PHY address	D	Loss of GPIO0 function, Loss of JTAG function, Wrong PHY address latching. No MDIO access / potentially conflicting PHY address	D	Loss of GPIO0 function, Loss of JTAG function, Wrong PHY address latching. No MDIO access / potentially conflicting PHY address	D	Loss of GPIO0 function, Loss of JTAG function, Wrong PHY address latching. No MDIO access / potentially conflicting PHY address	D	Loss of GPIO0 function, Loss of JTAG function, Wrong PHY address latching. No MDIO access / potentially conflicting PHY address	D	Loss of GPIO0 function, Loss of JTAG function, Wrong PHY address latching. No MDIO access / potentially conflicting PHY address
18	GPIO1 / TDI	D	19	RX_ER / GPIO2 / TDO / CONFIG1	Loss of GPIO0/1 function, Loss of JTAG function Communication possible if RX_ER not used in applcation.	D	Loss of JTAG / GPIO1 function.	D	Loss of JTAG / GPIO1 function.	D	Loss of JTAG / GPIO1 function.	D	Loss of JTAG / GPIO1 function.	D	Loss of JTAG / GPIO1 function.
19	RX_ER / GPIO2 / TDO / CONFIG1	B	20	RXD3 / GPIO3 / CONFIG2	Loss of JTAG. Erroneous pinstrap configuration read. Erroneous RX communication . Loss of GPIO functionality	D	Loss of JTAG. Erroneous pinstrap configuration read. Erroneous RX communication . Loss of GPIO functionality	D	Loss of JTAG. Erroneous pinstrap configuration read. Erroneous RX communication . Loss of GPIO functionality	D	Loss of JTAG. Erroneous pinstrap configuration read. Erroneous RX communication . Loss of GPIO functionality	D	Loss of JTAG. Erroneous pinstrap configuration read. Erroneous RX communication . Loss of GPIO functionality	D	Loss of JTAG. Erroneous pinstrap configuration read. Erroneous RX communication . Loss of GPIO functionality
20	RXD3 / GPIO3 / CONFIG2	B	21	RXD2 / GPIO4 / CONFIG3	Erroneous pinstrap configuration read.	B	Erroneous pinstrap configuration read.	B	Erroneous pinstrap configuration read.	B	Erroneous pinstrap configuration read.	B	Erroneous pinstrap configuration read.	B	Erroneous pinstrap configuration read.

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Table 40. TJA1121A*...continued*

Failure Classification

***A:** Damage to device / Bus may be affected*

***B:** No damage to device / No bus communication possible*

***C:** No damage to device / Bus communication possible / Corrupted node excluded from communication*

***D:** No damage to device / Bus communication possible / Reduced functionality of device*

— not applicable

Pin	Pin name	Class	Pin	Pin name	Mode	Class	Mode	Class	Mode	Class	Mode	Class	Mode	Class	Mode
					Erroneous RX communication in MII/RGMII mode. Loss of GPIO functionality		Erroneous RX communication in MII/RGMII mode. Loss of GPIO functionality		Erroneous RX communication in MII/RGMII mode. Loss of GPIO functionality		Erroneous RX communication in MII/RGMII mode. Loss of GPIO functionality		Erroneous RX communication in MII/RGMII mode. Loss of GPIO functionality		Erroneous RX communication in MII/RGMII mode. Loss of GPIO functionality
21	RXD2 / GPIO4 / CONFIG3	B	22	VDDIO	Erroneous pinstrap configuration read. Erroneous RX communication in MII/RGMII mode. Loss of GPIO functionality	B	Loss of GPIOs, MII/RMII/RGMII, SMI and IRQ access	B	Loss of GPIOs, MII/RMII/RGMII, SMI and IRQ access	—		B	Loss of GPIOs, MII/RMII/RGMII, SMI and IRQ access Potential corruption of Ics on 1.1V domain due to increased voltage (VDDIO)	B	Loss of GPIOS, MII/RMII/RGMII, SMI and IRQ access Potential corruption of Ics on 1.1V domain due to increased voltage (VDDIO)
22	VDDIO	B	23	RXD1 / CONFIG4	Erroneous pinstrap configuration read. Erroneous RX communication in MII/RGMII mode.	B	Erroneous pinstrap configuration read. Erroneous RX communication in MII/RGMII mode.	B	Erroneous pinstrap configuration read. Erroneous RX communication in MII/RGMII mode.	B	Erroneous pinstrap configuration read. Erroneous RX communication in MII/RGMII mode.	B	Erroneous pinstrap configuration read. Erroneous RX communication in MII/RGMII mode.	B	Erroneous pinstrap configuration read. Erroneous RX communication in MII/RGMII mode.
23	RXD1 / CONFIG4	B	24	RXD0 / CONFIG5	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.
24	RXD0 / CONFIG5	B	25	RX_CTL / CONFIG6	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.
25	RX_CTL / CONFIG6	B	26	RXC / GPIO8 / TX_TCLK	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.
26	RXC / GPIO8 / TX_TCLK	B	27	VDDIO	Erroneous pinstrap configuration read. Erroneous RX communication or loss of GPIO8 functionality (if used)	B	Erroneous pinstrap configuration read. Erroneous RX communication or loss of GPIO8 functionality (if used)	B	Erroneous pinstrap configuration read. Erroneous RX communication or loss of GPIO8 functionality (if used)	B	Erroneous pinstrap configuration read. Erroneous RX communication or loss of GPIO8 functionality (if used)	B	Erroneous pinstrap configuration read. Erroneous RX communication or loss of GPIO8 functionality (if used)	B	Erroneous pinstrap configuration read. Erroneous RX communication or loss of GPIO8 functionality (if used)
27	VDDIO	D	28	TX_ER / GPIO9 / TCK	Loss of JTAG function. Loss of GPIO function. Loss of TX_ER function, TX communication error if TX_ER pin is used in application	B	Loss of GPIOs, MII/RMII/RGMII, SMI and IRQ access	B	Loss of GPIOs, MII/RMII/RGMII, SMI and IRQ access	—		B	Loss of GPIOs, MII/RMII/RGMII, SMI and IRQ access Potential corruption of Ics on 1.1V domain due to increased voltage (VDDIO)	B	Loss of GPIOs, MII/RMII/RGMII, SMI and IRQ access Potential corruption of Ics on 1.1V domain due to increased voltage (VDDIO)
28	TX_ER / GPIO9 / TCK	B	29	TXC	Loss of GPIO9 function. Loss of JTAG function. Loss of TX communication	B	Loss of GPIO9 function. Loss of JTAG function. Loss of TX communication if TX_ER is used in application	B	Loss of GPIO9 function. Loss of JTAG function. Loss of TX communication if TX_ER is used in application	B	Loss of GPIO9 function. Loss of JTAG function. Loss of TX communication if TX_ER is used in application	B	Loss of GPIO9 function. Loss of JTAG function. Loss of TX communication if TX_ER is used in application	B	Loss of GPIO9 function. Loss of JTAG function. Loss of TX communication if TX_ER is used in application
29	TXC	B	30	TX_CTL	Loss of TX communication	B	Loss of TX communication	B	Loss of TX communication	B	Loss of TX communication	B	Loss of TX communication	B	Loss of TX communication
30	TX_CTL	B	31	TXD0	Loss of TX communication	B	Loss of TX communication	B	Loss of TX communication	B	Loss of TX communication	B	Loss of TX communication	B	Loss of TX communication

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Table 40. TJA1121A...*continued*

Failure Classification

A: *Damage to device / Bus may be affected*

B: *No damage to device / No bus communication possible*

C: *No damage to device / Bus communication possible / Corrupted node excluded from communication*

D: *No damage to device / Bus communication possible / Reduced functionality of device*

— not applicable

Pin	Pin name	Class	Pin	Pin name	Mode	Class	Mode	Class	Mode	Class	Mode	Class	Mode	Class	Mode
31	TXD0	B	32	TXD1	Loss of TX communication in MII/RGMII mode. Loss of GPIO10 and GPIO 11 functions	B	Loss of TX communication in MII/RGMII mode. Loss of GPIO10 function	B	Loss of TX communication in MII/RGMII mode. Loss of GPIO10 function	B	Loss of TX communication in MII/RGMII mode. Loss of GPIO10 function	B	Loss of TX communication in MII/RGMII mode. Loss of GPIO10 function	B	Loss of TX communication in MII/RGMII mode. Loss of GPIO10 function
32	TXD1	B	33	TXD2 / GPIO10	Loss of TX communication mode. Loss of GPIO10 functions	B	Loss of TX communication in MII/RGMII mode. Loss of GPIO11 function	B	Loss of TX communication in MII/RGMII mode. Loss of GPIO11 function	B	Loss of TX communication in MII/RGMII mode. Loss of GPIO11 function	B	Loss of TX communication in MII/RGMII mode. Loss of GPIO11 function	B	Loss of TX communication in MII/RGMII mode. Loss of GPIO11 function
33	TXD2 / GPIO10	B	34	TXD3 / GPIO11	Loss of TX communication mode (MII/RGMII mode). Loss of GPIOs	B	Loss of TX communication mode (MII/RGMII mode). Loss of GPIOs	B	Loss of TX communication mode (MII/RGMII mode). Loss of GPIOs	B	Loss of TX communication mode (MII/RGMII mode). Loss of GPIOs	B	Loss of TX communication mode (MII/RGMII mode). Loss of GPIOs	B	Loss of TX communication mode (MII/RGMII mode). Loss of GPIOs
34	TXD3 / GPIO11	B	35	VDDIO	Loss of TX communication mode (MII/RGMII mode). Loss of GPIOs	B	Loss of TX communication mode (MII/RGMII mode). Loss of GPIOs	B	Loss of TX communication mode (MII/RGMII mode). Loss of GPIOs	B	Loss of TX communication mode (MII/RGMII mode). Loss of GPIOs	B	Loss of TX communication mode (MII/RGMII mode). Loss of GPIOs	B	Loss of TX communication mode (MII/RGMII mode). Loss of GPIOs
35	VDDIO	D	36	MDIO	Loss of MDIO function	B	Loss of GPIOs, MII/RMII/RGMII, SMI and IRQ access	B	Loss of GPIOs, MII/RMII/RGMII, SMI and IRQ access	—		B	Loss of RX/TX communication Potential corruption of Ics on 1.1V domain due to increased voltage (VDDIO)	B	Loss of RX/RX communication Potential corruption of ICs on VDDIO domain due to increased voltage (VDDAO)
36	MDIO	D	1	MDC	Loss of MDIO function	D	Loss of MDIO function	D	Loss of MDIO function	D	Loss of MDIO function	D	Loss of MDIO function	D	Loss of MDIO function
37	GNDPAD														

Table 41. TJA1121B

Failure Classification

A: *Damage to device / Bus may be affected*

B: *No damage to device / No bus communication possible*

C: *No damage to device / Bus communication possible / Corrupted node excluded from communication*

D: *No damage to device / Bus communication possible / Reduced functionality of device*

— not applicable

Pin	Pin name	Class	Pin	Pin name	Mode	Class	Mode	Class	Mode	Class	Mode	Class	Mode	Class	Mode
1	MDC	D	2	INT_N	Loss of IRQ and MDIO functionality	D	Loss of IRQ and MDIO functionality	D	Loss of IRQ and MDIO functionality	D	Loss of IRQ and MDIO functionality	D	Loss of IRQ and MDIO functionality	D	Loss of IRQ and MDIO functionality
2	INT_N	A	3	VDD_ CORE	Limiting value of VDD_CORE exceeded	D	Loss of IRQ functionality. IRQ constantly asserted	D	Device in undervoltage	D	Loss of IRQ functionality. IRQ constantly deasserted	D	Loss of IRQ functionality	D	Loss of IRQ functionality. IRQ constantly deasserted. Potential damage to SoC depending on limiting values on IRQ pin.
3	VDD_ CORE	D	4	VDD_ CORE	Limiting value of VDD_CORE exceeded	B	Device in undervoltage	B	Device in undervoltage	A	Limiting value of VDD_CORE exceeded	—		A	Limiting value of VDD_CORE exceeded
4	VDD_ CORE	A	5	XO	Limiting value of XO exceeded	B	Device in undervoltage	B	Device in undervoltage	B	Device in undervoltage if VDDIO is not sufficiently high. Potential damage of IO pins due to increased VDDIO	B	Device in undervoltage	D	Reduced EMC / supply filtering performance
5	XO	B	6	XI / CLK_IN	No digital clock present	B	No digital clock available, no	B	No digital clock available, no	A	Limiting value of XO exceeded	B	No digital clock available, no	A	Limiting value of XO exceeded

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Table 41. TJA1121B*...continued*

Failure Classification

***A:** Damage to device / Bus may be affected*

***B:** No damage to device / No bus communication possible*

***C:** No damage to device / Bus communication possible / Corrupted node excluded from communication*

***D:** No damage to device / Bus communication possible / Reduced functionality of device*

— not applicable

Pin	Pin name	Class	Pin	Pin name	Mode	Class	Mode	Class	Mode	Class	Mode	Class	Mode	Class	Mode
							communication possible		communication possible				communication possible		
6	XI / CLK_IN	A	7	VREGA_ IN	Limiting value of XI exceeded	B	No digital clock available, no communication possible	B	No digital clock available, no communication possible	A	Limiting value of XO exceeded	B	No digital clock available, no communication possible	A	Limiting value of XO exceeded
7	VREGA_ IN	A	8	VDDA_RX	Limiting value of analog transmitter supply exceeded	B	No communication possible	B	No communication possible	B	If VDDIO is lower than 3.3V, no communication possible	B	No communication possible	D	Reduced EMC / supply filtering performance
8	VDDA_RX	A	9	VDDA_AO	Limiting value of analog transmitter supply exceeded	B	AFE not powered. No communication possible	B	AFE not powered. No communication possible	A	Limiting value of analog transmitter supply exceeded if VDDIO larget than 1.8V	B	AFE not powered. No communication possible	A	Limiting value of analog transmitter supply exceeded
9	VDDA_AO	D	10	RST_N	Loss of RST_N control via SoC. Internal POR and SW remains functional	B	Always on domain unpowered. No communication possible	B	Always on domain unpowered. No communication possible	B	Depends on level and state of VDDIO. For VDDIO lower than 3.3V, the always on domain is in undervoltage.	B	Always on domain is in undervoltage.	—	
10	RST_N	B	11	WAKE_ IN_OUT	Device will be kept in reset due to WAKE_IN_ OUT pulldown	B	Device is reset	D	Loss of RST_N control via SoC. Internal POR and SW remains functional	B	Depends on level and state of VDDIO. Device will be kept in reset for sufficiently low VDDIO levels.	B	Spurious device reset	D	Loss of RST_N pin function. SW reset still operational
11	WAKE_ IN_OUT	B	12	INH	If INH is used and connected to SBC/SMPS, potential loss of voltage supply. No effect, if INH and wake/sleep is not used in application	D	If wake/sleep not used, no effect. Loss of wake_in_ out mechanism	D	If wake/sleep not used, no effect. Loss of wake_in_ out mechanism and spurios wakeups if internal pull-down is not enabled	D	If wake/sleep not used, no effect. Possible corruption of SoC IO pins (VDDIO domain) via INH (always-on domain) Spurious wake/ wake_forwarding events.	D	If wake/sleep not used, no effect. Spurious wake/ wake_forwarding events.	D	If wake/sleep not used, no effect. Spurious wake/ wake_forwarding events.
12	INH	A	13	VDDA_ TRX	Limiting value of analog transmitter supply exceeded	D	No effect, if wake/ sleep not used. Early SMPS/SBC shutdown and loss of supply	D	No effect, if wake/ sleep not used. Early SMPS/SBC shutdown and loss of supply	D	No effect, if wake/ sleep not used. Potential loss of voltage supply if INH is connected to SBC/SMPS	D	No effect, if wake/ sleep not used. Potential loss of voltage supply if INH is connected to SBC/SMPS	D	No effect if wake/ sleep not used. Loss of INH shut-down functionality.
13	VDDA_ TRX	B	14	TRX_P	Loss of communication	A	Loss of communication. Device may get damaged	B	No communication	A	No communication. Limiting value of analog transmitter may get exceeded	B	No communication	A	No communication. Limiting value of analog transmitter may get exceeded
14	TRX_P	B	15	TRX_M	Loss of communication	B	Loss of communication	B	Loss of communication	B	Loss of communication	B	Loss of communication	B	Loss of communication
15	TRX_M	B	16	VDDA_ TRX	Loss of communication	B	Loss of communication	B	Loss of communication	B	Loss of communication	B	Loss of communication	B	Loss of communication
16	VDDA_ TRX	B	17	GPIO0 / TMS / CONFIG0	Loss of communication	A	Loss of communication. Device may get damaged	B	No communication	A	No communication. Limiting value of analog transmitter may get exceeded	B	No communication	A	No communication. Limiting value of analog transmitter may get exceeded
17	GPIO0 / TMS / CONFIG0	D	18	GPIO1 / TDI	Loss of GPIO0/1 function, Loss of JTAG function, Wrong PHY address latching. No MDIO access / potentially conflicting PHY address	D	Loss of GPIO0 function, Loss of JTAG function, Wrong PHY address latching. No MDIO access / potentially conflicting PHY address	D	Loss of GPIO0 function, Loss of JTAG function, Wrong PHY address latching. No MDIO access / potentially conflicting PHY address	D	Loss of GPIO0 function, Loss of JTAG function, Wrong PHY address latching. No MDIO access / potentially conflicting PHY address	D	Loss of GPIO0 function, Loss of JTAG function, Wrong PHY address latching. No MDIO access / potentially conflicting PHY address	D	Loss of GPIO0 function, Loss of JTAG function, Wrong PHY address latching. No MDIO access / potentially conflicting PHY address

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Table 41. TJA1121B*...continued*

Failure Classification

***A:** Damage to device / Bus may be affected*

***B:** No damage to device / No bus communication possible*

***C:** No damage to device / Bus communication possible / Corrupted node excluded from communication*

***D:** No damage to device / Bus communication possible / Reduced functionality of device*

— not applicable

Pin	Pin name	Class	Pin	Pin name	Mode	Class	Mode	Class	Mode	Class	Mode	Class	Mode	Class	Mode
18	GPIO1 / TDI	D	19	RX_ER / GPIO2 / TDO / CONFIG1	Loss of GPIO0/1 function, Loss of JTAG function Communication possible if RX_ER not used in applcation.	D	Loss of JTAG / GPIO1 function.	D	Loss of JTAG / GPIO1 function.	D	Loss of JTAG / GPIO1 function.	D	Loss of JTAG / GPIO1 function.	D	Loss of JTAG / GPIO1 function.
19	RX_ER / GPIO2 / TDO / CONFIG1	B	20	RXD3 / GPIO3 / CONFIG2	Loss of JTAG. Erroneous pinstrap configuration read. Erroneous RX communication . Loss of GPIO functionality	D	Loss of JTAG. Erroneous pinstrap configuration read. Erroneous RX communication . Loss of GPIO functionality	D	Loss of JTAG. Erroneous pinstrap configuration read. Erroneous RX communication . Loss of GPIO functionality	D	Loss of JTAG. Erroneous pinstrap configuration read. Erroneous RX communication . Loss of GPIO functionality	D	Loss of JTAG. Erroneous pinstrap configuration read. Erroneous RX communication . Loss of GPIO functionality	D	Loss of JTAG. Erroneous pinstrap configuration read. Erroneous RX communication . Loss of GPIO functionality
20	RXD3 / GPIO3 / CONFIG2	B	21	RXD2 / GPIO4 / CONFIG3	Erroneous pinstrap configuration read. Erroneous RX communication in MII/RGMII mode. Loss of GPIO functionality	B	Erroneous pinstrap configuration read. Erroneous RX communication in MII/RGMII mode. Loss of GPIO functionality	B	Erroneous pinstrap configuration read. Erroneous RX communication in MII/RGMII mode. Loss of GPIO functionality	B	Erroneous pinstrap configuration read. Erroneous RX communication in MII/RGMII mode. Loss of GPIO functionality	B	Erroneous pinstrap configuration read. Erroneous RX communication in MII/RGMII mode. Loss of GPIO functionality	B	Erroneous pinstrap configuration read. Erroneous RX communication in MII/RGMII mode. Loss of GPIO functionality
21	RXD2 / GPIO4 / CONFIG3	B	22	VDDIO	Erroneous pinstrap configuration read. Erroneous RX communication in MII/RGMII mode. Loss of GPIO functionality	B	Loss of GPIOs, MII/RMII/RGMII, SMI and IRQ access	B	Loss of GPIOs, MII/RMII/RGMII, SMI and IRQ access	—		B	Loss of GPIOs, MII/RMII/RGMII, SMI and IRQ access Potential corruption of Ics on 1.1V domain due to increased voltage (VDDIO)	B	Loss of GPIOS, MII/RMII/RGMII, SMI and IRQ access Potential corruption of Ics on 1.1V domain due to increased voltage (VDDIO)
22	VDDIO	B	23	RXD1 / CONFIG4	Erroneous pinstrap configuration read. Erroneous RX communication in MII/RGMII mode.	B	Erroneous pinstrap configuration read. Erroneous RX communication in MII/RGMII mode.	B	Erroneous pinstrap configuration read. Erroneous RX communication in MII/RGMII mode.	B	Erroneous pinstrap configuration read. Erroneous RX communication in MII/RGMII mode.	B	Erroneous pinstrap configuration read. Erroneous RX communication in MII/RGMII mode.	B	Erroneous pinstrap configuration read. Erroneous RX communication in MII/RGMII mode.
23	RXD1 / CONFIG4	B	24	RXD0 / CONFIG5	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.
24	RXD0 / CONFIG5	B	25	RX_CTL / CONFIG6	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.
25	RX_CTL / CONFIG6	B	26	RXC / GPIO8 / TX_TCLK	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Erroneous pinstrap configuration read. Erroneous RX communication.
26	RXC / GPIO8 / TX_TCLK	B	27	VDDIO	Erroneous pinstrap configuration read. Erroneous RX communication or loss of GPIO8 functionality (if used)	B	Erroneous pinstrap configuration read. Erroneous RX communication or loss of GPIO8 functionality (if used)	B	Erroneous pinstrap configuration read. Erroneous RX communication or loss of GPIO8 functionality (if used)	B	Erroneous pinstrap configuration read. Erroneous RX communication or loss of GPIO8 functionality (if used)	B	Erroneous pinstrap configuration read. Erroneous RX communication or loss of GPIO8 functionality (if used)	B	Erroneous pinstrap configuration read. Erroneous RX communication or loss of GPIO8 functionality (if used)

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Table 41. TJA1121B*...continued*

Failure Classification

***A:** Damage to device / Bus may be affected*

***B:** No damage to device / No bus communication possible*

***C:** No damage to device / Bus communication possible / Corrupted node excluded from communication*

***D:** No damage to device / Bus communication possible / Reduced functionality of device*

— not applicable

Pin	Pin name	Class	Pin	Pin name	Mode	Class	Mode	Class	Mode	Class	Mode	Class	Mode	Class	Mode
27	VDDIO	D	28	GPIO9 / TCK	Loss of JTAG function. Loss of GPIO function. Loss of TX_ER function, TX communication error if TX_ER pin is used in application	B	Loss of GPIOs, MII/RMII/RGMII, SMI and IRQ access	B	Loss of GPIOs, MII/RMII/RGMII, SMI and IRQ access	—		B	Loss of GPIOs, MII/RMII/RGMII, SMI and IRQ access Potential corruption of Ics on 1.1V domain due to increased voltage (VDDIO)	B	Loss of GPIOs, MII/RMII/RGMII, SMI and IRQ access Potential corruption of Ics on 1.1V domain due to increased voltage (VDDIO)
28	GPIO9 / TCK	B	29	VDDA_SO	Loss of JTAG function. Loss of GPIO9 function. Loss of SGMII connectivity	D	Loss of JTAG and GPIO9 function	D	Loss of JTAG and GPIO9 function	D	Loss of JTAG and GPIO9 function	D	Loss of JTAG and GPIO9 function	D	Loss of JTAG and GPIO9 function
29	VDDA_SO	A	30	SON	SGMII PMA may get damaged	B	Loss of SGMII connectivity	B	Loss of SGMII connectivity	A	Limiting value of SGMII supply exceed	—		A	Limiting value of SGMII supply exceed
30	SON	B	31	SOP	Loss of RX communication	B	Loss of RX connectivity	B	Loss of RX connectivity	A	Loss of RX connectivity; SGMII PMA may get damaged	A	Loss of RX connectivity; SGMII PMA may get damaged	A	Loss of RX connectivity; SGMII PMA may get damaged
31	SOP	B	32	SIP	Loss of RX/TX communication	B	Loss of RX connectivity	B	Loss of RX connectivity	A	Loss of RX connectivity; SGMII PMA may get damaged	A	Loss of RX connectivity; SGMII PMA may get damaged	A	Loss of RX connectivity; SGMII PMA may get damaged
32	SIP	B	33	SIN	Loss of TX communication	B	Loss of TX communication	B	Loss of TX communication	A	Loss of TX connectivity; SGMII PMA may get damaged	A	Loss of TX connectivity; SGMII PMA may get damaged	A	Loss of TX connectivity; SGMII PMA may get damaged
33	SIN	A	34	VDDA_SI	SGMII PMA may get damaged	B	Loss of TX communication	B	Loss of TX communication	A	Loss of TX connectivity; SGMII PMA may get damaged	A	Loss of TX connectivity; SGMII PMA may get damaged	A	Loss of TX connectivity; SGMII PMA may get damaged
34	VDDA_SI	A	35	VDDIO	SGMII PMA may get damaged	B	Loss of SGMII connectivity	B	Loss of SGMII connectivity	A	Limiting value of SGMII supply exceed	—		A	Limiting value of SGMII supply exceed
35	VDDIO	B	26	RXC / GPIO8 / TX_TCLK	Erroneous pinstrap configuration read. Erroneous RX communication.	B	Loss of SMI and IRQ access	B	Loss of RX communication	—		B	Loss of RX communication Potential corruption of Ics on 1.1V domain due to increased voltage (VDDIO)	B	Loss of RX communication Potential corruption of Ics on 1.1V domain due to increased voltage (VDDIO)
36	MDIO	D	1	MDC	Loss of MDIO function	D	Loss of MDIO function	D	Loss of MDIO function	D	Loss of MDIO function	D	Loss of MDIO function	D	Loss of MDIO function
37	GNDPAD														

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8 PCB Checklist

Table 42. PCB Checklist - TJA1121A

Pin name	Pin number	TJA1121 Requirement
		Check pinning and correct pin numbers. Pay attention to the differences between the A and B variants
MDC	1	MDC pin must be connected to the MDC output of the MDIO Leader device (typically to the µC). Provision optional series resistor at the driver (close the µC) to improve EMC performance. The impedance of the series resistor plus the output impedance of the MDC driver shall match the impedance of the PCB trace.
		If MDC is distributed to multiple PHYs, keep the branches and stubs as short as possible.
INT_N	2	If INT_N is used, connect pullup resistor of (for example, 10 kΩ) to VDDIO. This can be integrated pullup of the µC. The value of the pullup resistor should be chosen such that a proper high/low level is ensured.
		If not used it can be left open.
VDD_CORE	3	Connect a 100 nF and a 1 µF ceramic capacitor between this pin and GND close to the pin. Add a ferrite bead for filtering. See Section 3.5 for details.
VDD_CORE	4	
XO	5	If this pin is not used, it can be left open.
		Make sure the pin-strapping configuration is matching the clocking concept. The device might be damaged when high voltage (3.3 V) is applied while low voltage mode (differential/XO mode) is selected If a 25 MHz external crystal (±100ppm) is used, follow the crystal supplier recommendations. Provision for optional series resistor.
XI	6	If this pin is not used, it should be shorted to GND to avoid noise input. Provision for optional series resistor.
VREGA_IN	7	Connect a 100 nF and a 1 µF ceramic capacitor between this pin and GND close to the pin. Add a ferrite bead for filtering. See Section 3.5 for details.
VDDA_RX	8	Connect a 1 µF ceramic capacitor.
VDDA_AO	9	Connect a 100 nF ceramic capacitor between this pin and GND close to the pin.
RST_N	10	Connect to GPIO of the µC. A pullup resistor of about 10 kΩ to VDD_AO possible, but not necessary.
WAKE_IN_OUT	11	If used as a wake-up output, the pin has a high-side switch to VDDA_AO level when activated, otherwise it is high-Z.
		If not used, connect a pulldown resistor to ensure a proper low level.
INH	12	If used, connect INH to voltage regulators within the ECU (enable signal) via an external or embedded pulldown of 5kΩ - 20kΩ.
		If not used, INH can be left open.
VDDA_TRX	13, 16	The two VDDA_TRX pins should be connected together on the PCB.
TRX_P	14	The recommended circuit can be found in Section 3.4.
TRX_M	15	

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Table 42. PCB Checklist - TJA1121A *...continued*

Pin name	Pin number	TJA1121 Requirement
RXD3	20	No external termination resistors are necessary. 0 ohm resistors for fine tuning recommended.
RXD2	21	
RXD1	23	
RXD0	24	
RXD0	24	
RX_CTL	25	
RXC	26	
TXC	29	
VDDIO	22, 27, 35	The three VDDIO pins should be connected together on the PCB.
TXD0	31	No external resistors are necessary. 0 ohm resistors recommended for fine tuning
TXD1	32	
TXD2	33	
TXD3	34	
TX_CTL	3	
MDIO	36	Connect to MDIO pin of µC with a pullup resistor of 1.5 kΩ to 10 kΩ to $V_{DD(IO)}$, the pullup can be also within the µC. Make sure that the pullup resistor is available only once at the same MDIO bus.
GNDPAD	37	The exposed die pad shall be soldered to GND with a low impedance. It is recommended to have a ring of GND vias for optimized EMC behavior.

Table 43. PCB Checklist - TJA1121B

Pin name	Pin number	TJA1121 Requirement
		Check pinning and correct pin numbers. Pay attention to the differences between the A and B variants
MDC	1	MDC pin must be connected to the MDC output of the MDIO Leader device (typically to the µC). Provision an optional series resistor at the driver (close the µC) to improve EMC performance. The impedance of the series resistor plus the output impedance of the MDC driver shall match the impedance of the PCB trace.
		If MDC is distributed to multiple PHYs, keep the branches and stubs as short as possible.
INT_N	2	If INT_N is used, connect a pullup resistor of (for example, 10 kΩ) to VDDIO. This can be integrated pullup of the µC. The value of the pullup resistor should be chosen such that a proper high/low level is ensured.
		If not used it can be left open.
VDD_CORE	3	Connect a 100 nF and a 1 µF ceramic capacitor between this pin and GND close to the pin. Add a ferrite bead for filtering. See Section 3.5 for details.
VDD_CORE	4	
XO	5	If this pin is not used, it can be left open.

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Table 43. PCB Checklist - TJA1121B...*continued*

Pin name	Pin number	TJA1121 Requirement
		Make sure the pin-strapping configuration matches the clocking concept. The device may be damaged when a high voltage (3.3 V) is applied while low voltage mode (differential/XO mode) is selected If a 25 MHz external crystal (±100ppm) is used, follow the crystal supplier recommendations. Provision for optional series resistor.
XI	6	If this pin is not used, it should be shorted to GND to avoid noise input. Provision for optional series resistor.
VREGA_IN	7	Connect a 100 nF and a 1 µF ceramic capacitor between this pin and GND close to the pin. Add a ferrite bead for filtering. See Section 3.5 for details.
VDDA_RX	8	Connect a 1 µF ceramic capacitor.
VDDA_AO	9	Connect a 100 nF ceramic capacitor between this pin and GND close to the pin.
RST_N	10	Connect to GPIO of the µC. A pullup resistor of about 10 kΩ to VDD_AO is possible, but not necessary.
WAKE_IN_OUT	11	If used as a wake-up output, the pin has a high-side switch to VDDA_AO level when activated, otherwise it is high-Z.
		If not used, connect a pulldown resistor to ensure a proper low level.
INH	12	If used, connect INH to voltage regulators to enable signal within the ECU. A pulldown resistor of 5 kΩ - 20 kΩ is required externally. If not used, INH can be left open.
VDDA_TRX	13, 16	The two VDDA_TRX pins should be connected together on the PCB.
TRX_P	14	The recommended circuit can be found in Section 3.4.
TRX_M	15	
VDDIO	22, 27, 35	The three VDDIO pins should be connected together on the PCB.
VDDA_SO	29	Connect together with VDDA_SI (pin 34) on the PCB.
SON	30	Can be swapped with SOP (pin 31) for improved routing.
SOP	31	Can be swapped with SON (pin 30) for improved routing.
SIP	32	Can be swapped with SIN (pin 33) for improved routing.
SIN	33	Can be swapped with SIP (pin 32) for improved routing.
VDDA_SI	34	Connect together with VDDA_SO (pin 29) on the PCB.
MDIO	36	Connect to MDIO pin of µC with a pullup resistor of 1.5 kΩ to 10 kΩ to $V_{DD(IO)}$, the pullup can be also within the µC. Make sure the pullup resistor is available only once at the same MDIO bus.
GNDPAD	37	The exposed die pad shall be soldered to GND with a low impedance. It is recommended to have a ring of GND vias for optimized EMC behavior.

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9 Abbreviations

Table 44. Abbreviations

Term	Definition
CDM	Charge-device model
CDR	Clock-data recovery
CMC	Common-mode choke
EMC	Electromagnetic compatibility
ESD	Electrostatic discharge
FCS	Frame Check Sequence
HBM	Human-body model
MACsec	Media Access Control Security
MDI	Media Dependent Interface
MDIO	Management Data Input/Output
MII	Media Independent Interface
PCS	Physical Coding Sublayer
PHY	Physical Layer
PMA	Physical Medium Attachment
RGMII	Reduced Gigabit Media Independent Interface
RMII	Reduced Media Independent Interface
SGMII	Serial Gigabit Media Independent Interface
TDR	Time Domain Reflectometer

10 References

[1] TJA1121 data sheet, secure files at nxp.com/TJA1121
[2] AH1806 PCB design guide, Secure Files at nxp.com/TJA1121
[3] IEC 62228-5 2019-09 Integrated circuits - EMC evaluation of transceivers - Part 5: Ethernet transceivers

11 Revision history

Table 45. Revision history

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